Human and mouse TNF-alpha binders

The selected 20: candidate histories and bound structures

Main text | 19 October 2026

Twenty candidates | Human and mouse TNF-alpha | Top and side structural views

We prioritize 20 exact protein sequences for human and mouse TNF-alpha, seeking interaction at pH 7.4 and release at pH 6.0. Each selected molecule descends from hotspot-conditioned Foundry/RFdiffusion3 generation followed by Proton-PottsMPNN sequence redesign. Seven are GS-linked two-domain fusions, four carry subsequent expert-specified substitutions, and nine retain their selected Proton-PottsMPNN sequence. We reconstruct the recorded steps for every candidate and show top and side views of all 120 selected human/mouse complexes predicted by Boltz2, OpenDDE, and AlphaFold3. Scores support computational prioritization.

Documented campaign effort: at least **49,796 design/proposal attempts**; **47,962 distinct exact sequences** in the audited outputs. The attempt count is a lower bound; section 1.1 defines its scope.

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1 Introduction

The objective is a protein binder that recognizes soluble human TNF-alpha, cross-reacts with mouse TNF-alpha, and preferentially interacts at pH 7.4 rather than pH 6.0. We used complementary structural and sequence evidence to prioritize molecules for experimental assessment. The present main text is restricted to the 20 exact sequences in the October 19 ranked review CSV; its order is preserved throughout.

Retained campaign records document at least **49,796 design/proposal attempts**, with **48,081 sequence-output records representing 47,962 distinct exact sequences**. This is a documented lower bound across initial generation, redesign, expert mutation and fusion routes; it excludes refold replicas and target-only sampling models. The selected 20 are the focus of this report. Section 1.1 defines the count and its scope.

The initial strategy used AFSample3 conformational exploration of human and mouse TNF-alpha, including exploratory six-protomer systems, to guide hotspot choice. A target-only six-protomer input and a TNF trimer with three candidate copies are distinct molecular systems. The retained record supports the researcher-described role of conformational sampling in hotspot selection, but does not supply a complete sample-to-hotspot causal map for every candidate. The side and top epitope depictions are therefore retained as design context, without assigning an unrecorded sampled conformation to an individual molecule.

Hotspot-conditioned Foundry/RFdiffusion3 designs were refolded against both species and screened for interface confidence and structural preservation. Proton-PottsMPNN then proposed and optimized sequences on paired parent backbones for the two-pH objective. Selected descendants were either retained intact, changed by explicit expert substitutions, or combined as exact domains joined by geometry-guided GS linkers. Later three-copy predictions and copy-specific pH assessments tested the resulting exact sequences. DeltaForge KD predictions assess one complete candidate monomer against the human TNF-alpha trimer and, separately, against the mouse TNF-alpha trimer.

Each candidate section follows this sequence of evidence: original target and hotspot input; RFD3 source identity and sequence extraction; parent refolding and admission; Proton-PottsMPNN proposal and optimization; any subsequent mutation or fusion; downstream refolding and scoring; and current model-specific validation. Recorded settings and optimization trajectories support the candidate histories.

1.1 Campaign design effort

The documented minimum is **49,796 design/proposal attempts**. Completed records contain **48,081 sequence outputs** and **47,962 distinct exact sequences** across the audited branches. Attempt records include rejected proposals where their logs survive; sequence-output records count completed design outcomes, including proposals not advanced to refolding. Identical sequences across runs count once in the distinct-sequence total.

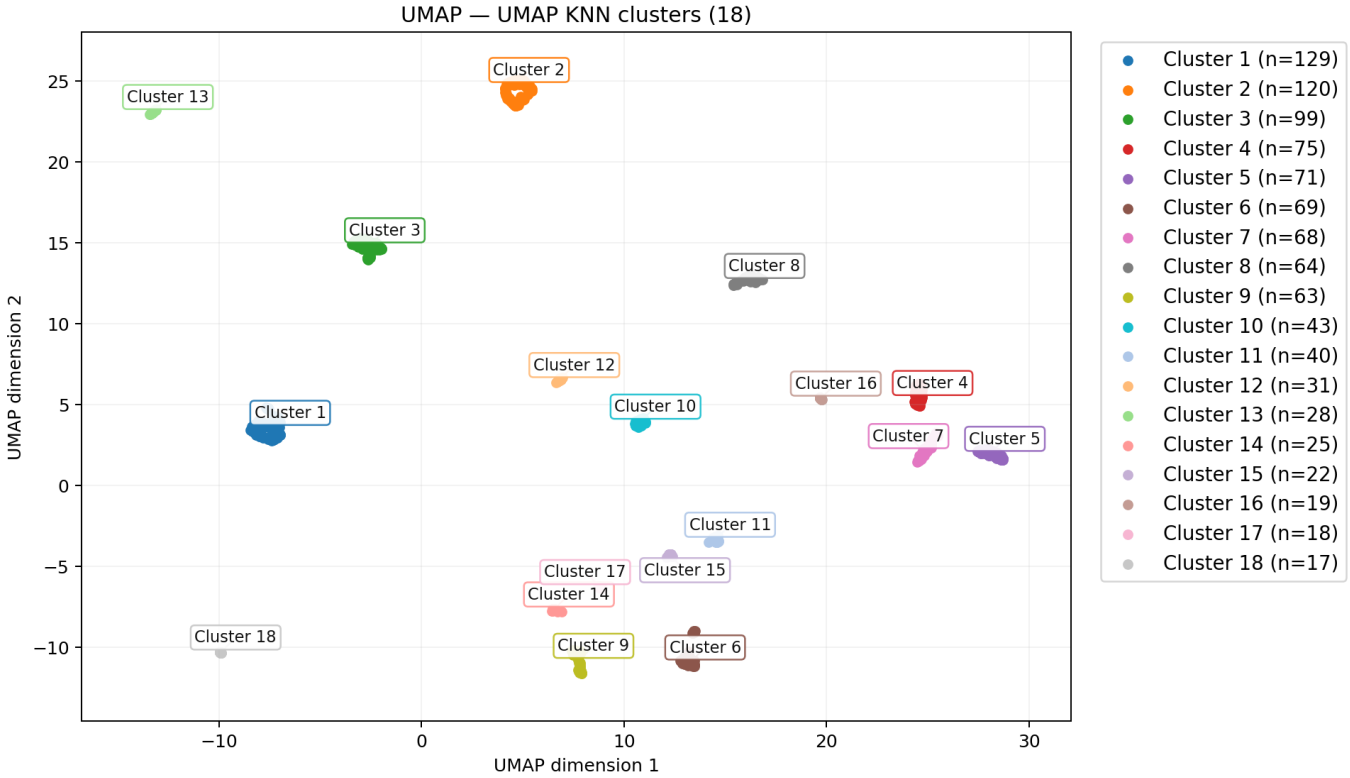
Generation or redesign branch	Sequence outputs	Attempts / minimum
RFD3 initial assessed designs	32,488	>= 32,488
Proton-Potts pilot 1	8	9
Proton-Potts pilot 2	16	17
Proton-Potts production round 1	6,400	7,108
Proton-Potts production round 2	3,200	3,732
Targeted paired redesign	64	65
Targeted conservative redesign	32	504
AP Novo-derived Proton-Potts pool	640	>= 640
Additional AP Novo-derived exact redesigns	160	>= 160
AP Novo pilot	8	8
AP Novo initial production	800	800
AP Novo extension	3,200	3,200
AP Novo top-epitope production	1,000	1,000
Expert mutants	16	16
Geometry-guided GS fusions	47	47
RF optimization final outputs	2	>= 2
Documented total	48,081	>= 49,796

The RFD3 contribution is the 32,488 completed candidates in the paired structural assessment; it is not a complete original generation-attempt ledger. The main Proton-PottsMPNN rounds record 7,108 and 3,732 proposal attempts, producing pools of 6,400 and 3,200 accepted sequences; 4,000 and 2,000, respectively, advanced to paired refolding. The additional sequence pools and later design routes are listed separately above.

The overall attempt count remains a lower bound because the complete original RFD3 attempt history and rejected-proposal logs for two downstream pools are unavailable. Only the two terminal RF optimization outputs are counted, rather than internal iterations. Recovery/intake copies, repeated preparation, the unchanged expert control, target-only conformational samples and structure-prediction replicas are excluded. These campaign-wide counts do not imply that every branch contributed to the selected 20.

1.2 UMAP: projection used for cluster selection

Original 5TSW hexamer projection used to select the 18-cluster partition. A symmetrized 15-neighbor connectivity graph constrained Ward agglomeration in the two-dimensional UMAP space; the maximum Euclidean silhouette among 2-20 clusters selected 18 (silhouette 0.902220). UMAP used 15 neighbors, min_dist 0.1 and random seed 42 [19,21,22].

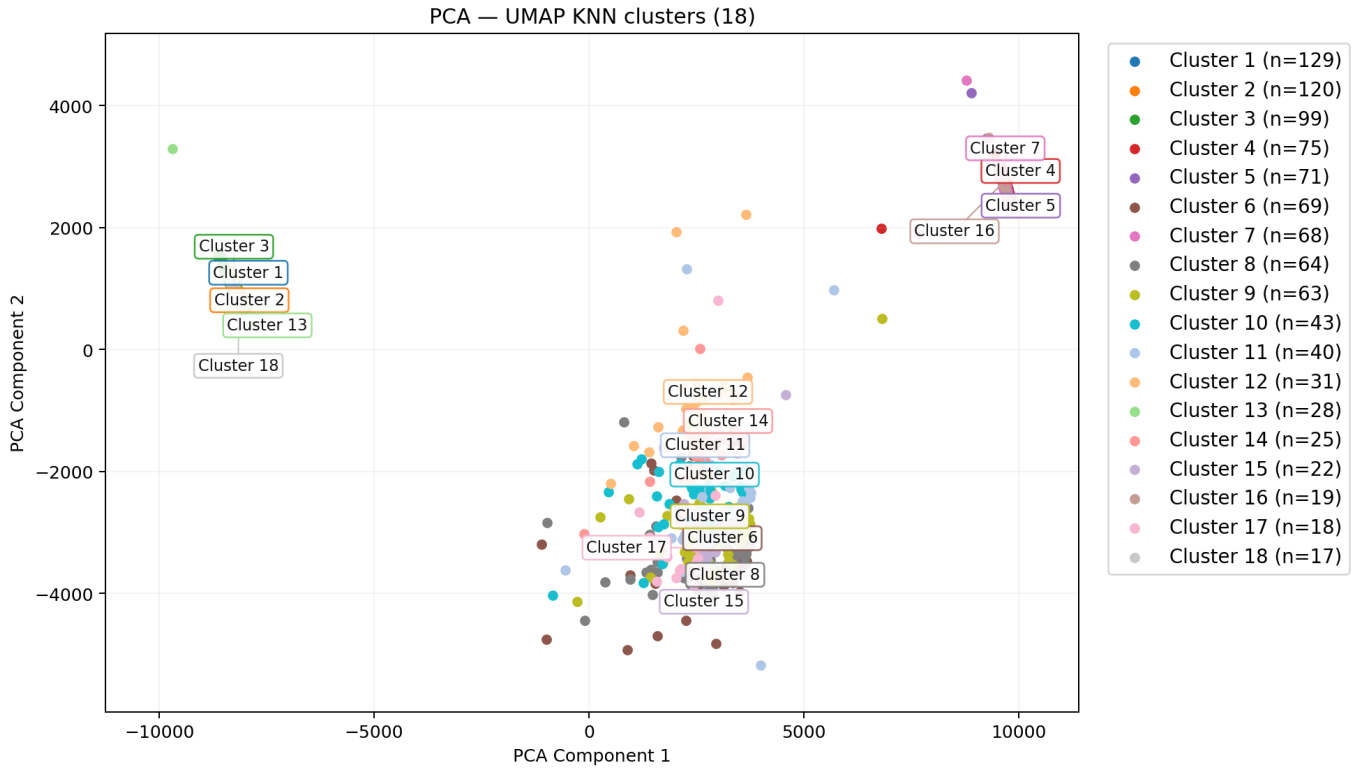


Original UMAP projection, retaining all 18 labels, colors and the source legend. The plotted population contains 1,001 entries; cluster 8 includes one additional top-level output. Category sample counts on the following pages omit that entry. The existing PNG is unchanged.

The head-to-head, side-to-side and tail-to-tail categories below group these original clusters by representative trimer-pair geometry. They do not replace the 18 cluster assignments.

1.3 PCA: the same cluster assignments

Original PCA projection of the same shared-residue C-alpha distance vectors. Cluster labels and colors come from the UMAP/Ward partition; PCA was not reclustered independently. PCA used the scikit-learn implementation [18].

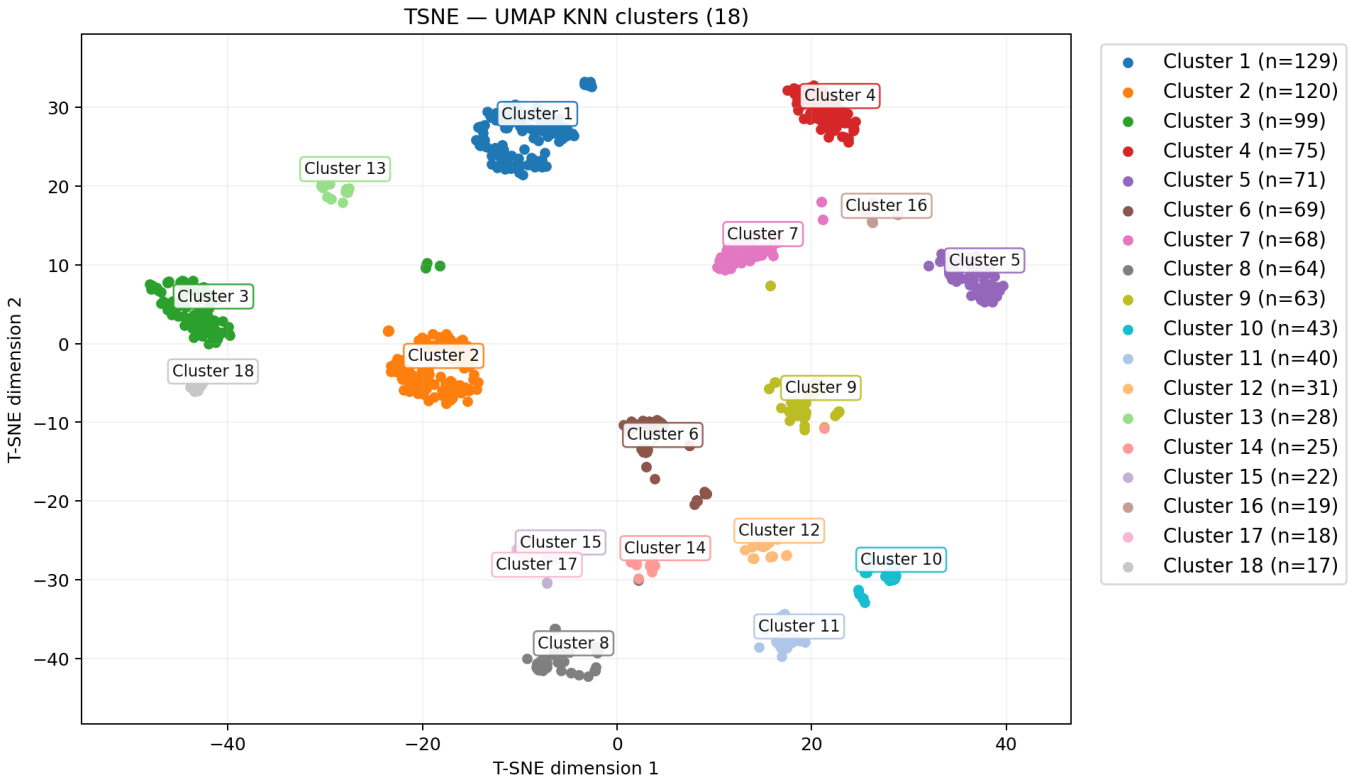


Original PCA projection, retaining all 18 labels, colors and the source legend. The plotted population contains 1,001 entries; cluster 8 includes one additional top-level output. Category sample counts on the following pages omit that entry. The existing PNG is unchanged.

The head-to-head, side-to-side and tail-to-tail categories below group these original clusters by representative trimer-pair geometry. They do not replace the 18 cluster assignments.

1.4 t-SNE: the same cluster assignments

Original t-SNE projection of the same shared-residue C-alpha distance vectors, using the Python/scikit-learn implementation with perplexity 30 and random seed 42 [18,20]. Labels and colors retain the UMAP/Ward partition.



Original TSNE projection, retaining all 18 labels, colors and the source legend. The plotted population contains 1,001 entries; cluster 8 includes one additional top-level output. Category sample counts on the following pages omit that entry. The existing PNG is unchanged.

The head-to-head, side-to-side and tail-to-tail categories below group these original clusters by representative trimer-pair geometry. They do not replace the 18 cluster assignments.

1.5 AFSample3 hexamer interaction classes

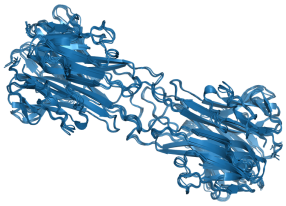
The opening ensembles reuse the existing 5TSW-derived human TNF-alpha hexamer clustering: six 150-residue M3S-construct chains, 18 clusters, and five displayed representatives per cluster. This exploratory construct is distinct from the later human and mouse binder-design constructs. Structural source: RCSB 5TSW.

The head is the face containing the top epitope; the tail is the opposite face. Head-to-head is the project's canonical top-to-top orientation. Side-to-side brings the lateral surfaces together, whereas tail-to-tail brings the faces opposite the top epitope together. The categories describe sampled trimer-pair geometries.

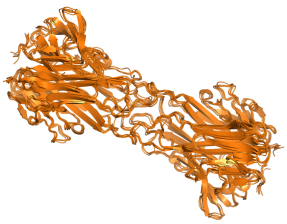
Interaction class	Clusters	Samples / 1,000
head-to-head	1, 2, 3, 13, 18	393
side-to-side	6, 8, 9, 10, 11, 12, 14, 15, 17	374
tail-to-tail	4, 5, 7, 16	233

The original clustering contains 1,001 entries: 1,000 seed/sample models and one additional top-level output in cluster 8. Counts here exclude that extra output while preserving every cluster assignment. All 90 displayed representatives agree with their cluster's geometric category. Original ensemble PNGs are reused unchanged.

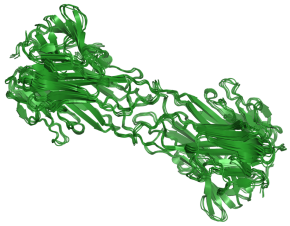
Cluster 1 | 129 sampled models



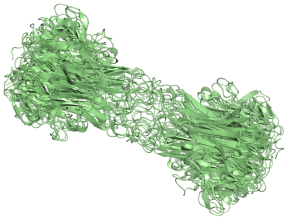
Cluster 2 | 120 sampled models



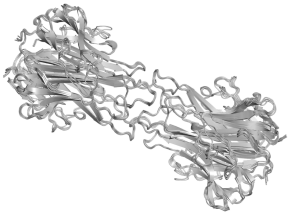
Cluster 3 | 99 sampled models



Cluster 13 | 28 sampled models



Cluster 18 | 17 sampled models

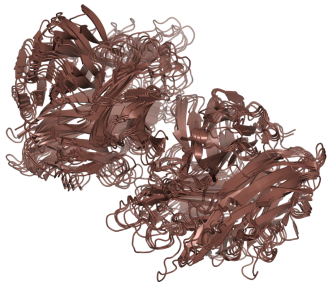


Opening ensemble plate A. Head-to-head clusters 1, 2, 3, 13 and 18. Each original PNG overlays the five selected representative models; the displayed populations are selections from a cluster, rather than the full cluster.

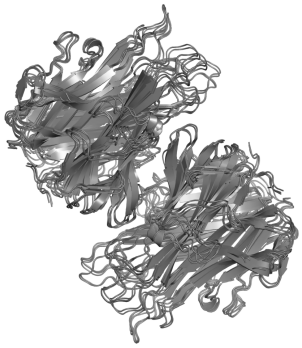
1.6 Side-to-side ensembles

Lateral trimer surfaces contact one another; the top-epitope faces are not opposed. Original cluster colors and ensemble depictions are retained.

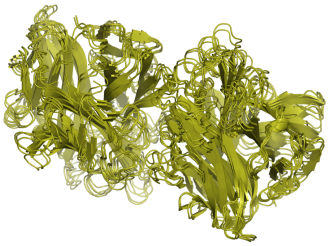
Cluster 6 | 69 sampled models



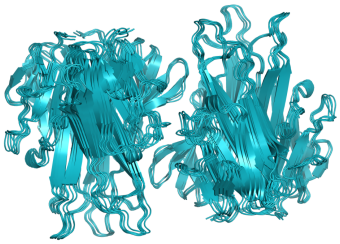
Cluster 8 | 63 sampled models



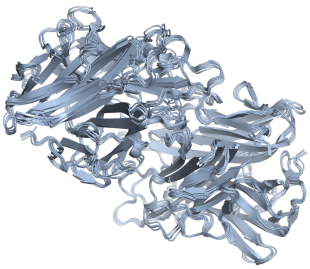
Cluster 9 | 63 sampled models



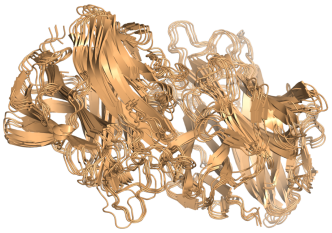
Cluster 10 | 43 sampled models



Cluster 11 | 40 sampled models



Cluster 12 | 31 sampled models

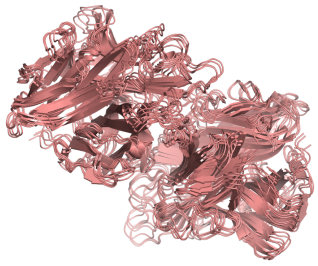


Opening ensemble plate B1. Side-to-side representative ensembles. Sample counts exclude the additional top-level output in cluster 8.

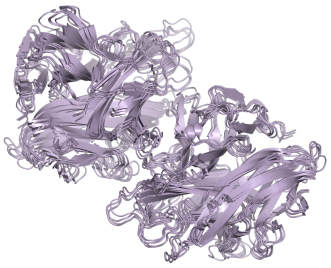
1.6 Side-to-side ensembles - continued

Lateral trimer surfaces contact one another; the top-epitope faces are not opposed. Original cluster colors and ensemble depictions are retained.

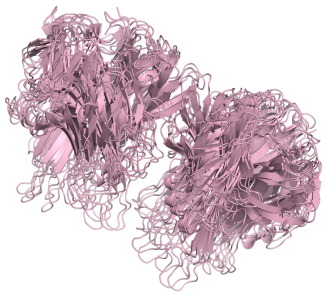
Cluster 14 | 25 sampled models



Cluster 15 | 22 sampled models



Cluster 17 | 18 sampled models

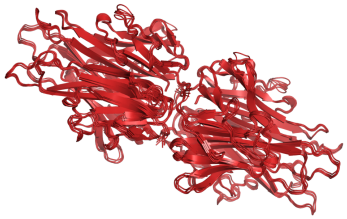


Opening ensemble plate B2. Side-to-side representative ensembles. Sample counts exclude the additional top-level output in cluster 8.

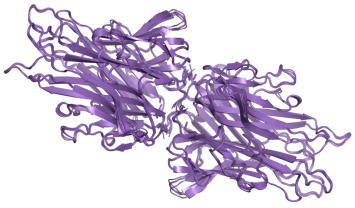
1.7 Tail-to-tail ensembles

The faces opposite the top epitope meet. The top-epitope faces point away from the intertrimer contact, distinguishing these ensembles from the head-to-head class.

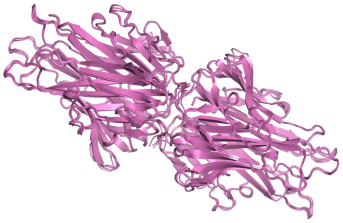
Cluster 4 | 75 sampled models



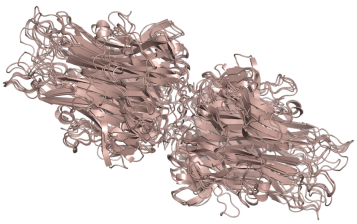
Cluster 5 | 71 sampled models



Cluster 7 | 68 sampled models



Cluster 16 | 19 sampled models

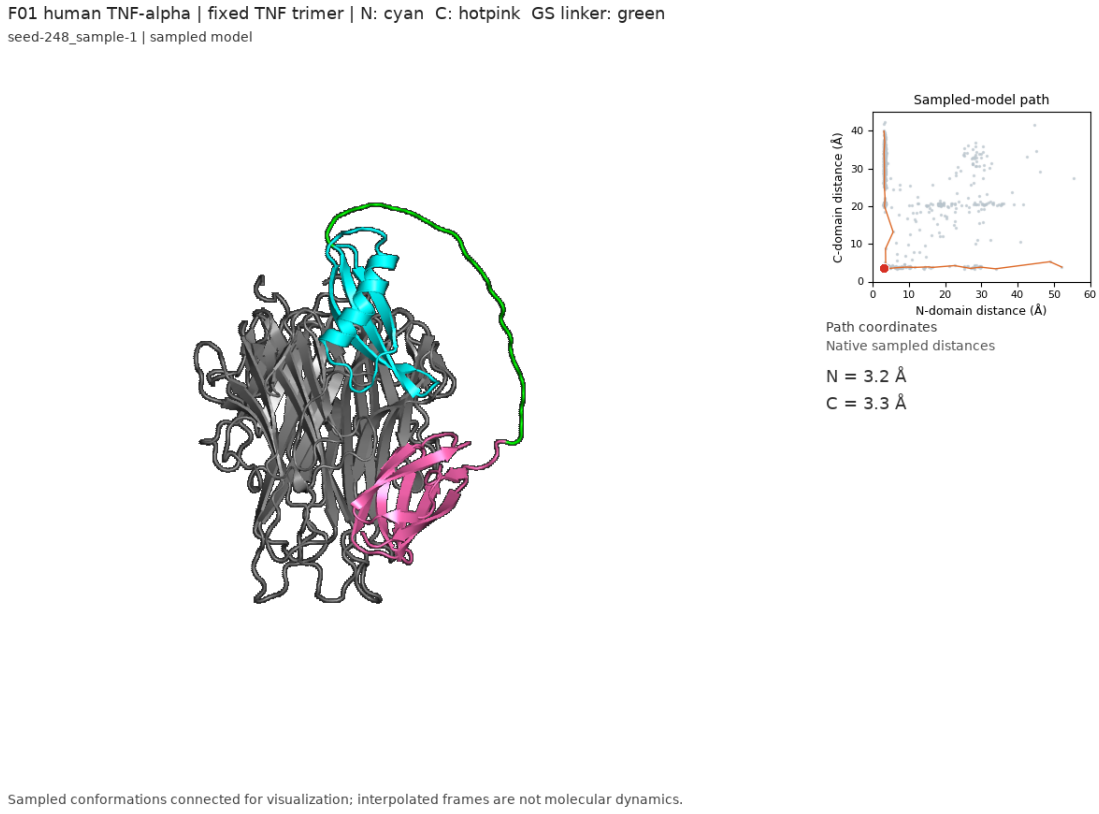


Opening ensemble plate C. Tail-to-tail clusters 4, 5, 7 and 16. Existing PNGs show the five selected representatives from each original cluster.

1.8 F01 AFSample3 sampled-state animation

For the selected two-domain candidate F01, AFSample3 sampling also explores domain-linker-domain configurations around human TNF-alpha. A distance analysis contains 1,000 native models: 900 single-binder models and 100 retained three-binder models, with one linked binder analyzed per model. The animation follows 23 selected anchors: 21 from the single-binder cohort and two from the three-binder cohort. One linked binder and one fixed target trimer are displayed throughout.

The sampled anchors illustrate N-terminal-domain proximity, proximity of both domains, and C-terminal-domain proximity to their corresponding parent-defined TNF pockets. Each pocket contains target residues within 5 angstrom of its original bound parent: 22 residues for the N-terminal domain and 24 for the C-terminal domain. The plotted distance is the mean over pocket residues of their minimum heavy-atom distance to the corresponding domain; linker atoms are excluded.



F01 sampled-state animation. Existing middle-frame PNG from the regenerated zoomed movie, showing a native sampled anchor. Gray: fixed human TNF trimer; cyan: N-terminal domain; green: GS linker; hot pink: C-terminal domain. The source image and palette are retained unchanged.

All anchors use target-only rigid alignment to the same display trimer. The 15.73-second movie has 236 logical frames at 15 fps. Smooth transitions interpolate domain rotations, translations and internal coordinates, with linker endpoint correction. These transitions visualize the sampled configurations; they are not additional predictions or a molecular-dynamics trajectory. Native anchor measurements and interpolated path coordinates are distinguished in the movie inset. The animation supplements the selected F01 models and exact-sequence scores.

View the F01 animation and measurements, GIF, or MP4.

2 Shared methods and interpretation

Structural features, projections and representative selection

The Structure_Clustering workflow aligns sequences and retains shared C-alpha positions. Each structure is encoded by the upper triangle of its C-alpha distance matrix, excluding the diagonal. The same distance-vector features feed two-dimensional scikit-learn PCA [18], umap-learn UMAP [19], and Python/scikit-learn t-SNE [18,20]. Original run settings are retained; current installed package versions are not substituted for historical generating versions.

The original UMAP partition used 15-neighbor graph connectivity and Ward agglomeration [21]. Euclidean silhouette [22] compared partitions of 2-20 clusters and selected 18 at 0.902220. The same source-path assignments and cluster palette label PCA, UMAP and t-SNE. Existing embeddings, assignments and PNGs are reused rather than recomputed. Cluster membership is a structural grouping, not an equilibrium population estimate.

Each cluster displays its five highest-ipTM representatives. The original structural medoid minimizes mean pairwise whole-complex C-alpha RMSD over all cluster members with fixed chain/residue correspondence and no outlier rejection; it is not the center of the two-dimensional projection. Representative states were rigidly aligned to that structural reference for the saved ensemble PNGs. The new common orientation for candidate complexes uses a proper least-squares rigid fit [23]. The original PNGs and their cameras are retained for the opening hexamer ensembles.

2.1 Target constructs, hotspots, and source identity

The recorded paired constructs contain three human chains of 152 residues or three mouse chains of 148 residues, labeled A/B/C. Original source structures are 1TNF-associated human and 2TNF-associated mouse coordinates. Hotspots use source author numbering, not automatically the renumbered predictor input. The candidate histories name the exact side/top epitope cohort and author-numbered residues. Human soluble-domain reference and challenge sequence identities require separate review; the report preserves the actual modeled constructs rather than silently substituting another sequence.

Foundry/RFD3 source sequences were exported from native flat-CIF coordinate residue identities. The nine source scaffolds underlying the selected set have retained output specifications and design inputs. These provide the target contigs, fixed-target settings, hotspot atom selections, and source residue numbering used in each candidate history.

The human target contig is A6-157,/0,B6-157,/0,C6-157,/0,60-100. The mouse target contig is A9-72,A74-157,/0,B9-72,B74-157,/0,C9-72,C74-157,/0,60-100, retaining the coordinate-resolved gap at author residue 73. Both specify fixed target atoms, select_unfixed_sequence=false, and hotspot-based orientation, with one separate 60-100-residue de novo chain. Input parsing retains covalent/disulfide annotations and does not alter formal charges. Atom-level hotspot selectors appear in the candidate histories.

The side epitope is the initial hotspot set; the top epitope is the subsequently selected hotspot set. These names describe the chosen target surface regions. P and RP identify two parent-selection rounds, rather than the epitope itself. The source specifications record no fixed RFD3 seed; the recorded downstream redesign seeds are reported separately.

For the opening hexamer categories, sequence correspondence transfers the top-epitope axis to each trimer. The two trimer partitions are identified by target-only fits to a common trimer reference. The head axis points from each trimer center toward the mean C-alpha position of source-human residues 103, 104, 107, 109 and 110, using their sequence-mapped equivalents. Head-to-head requires both head-axis projections toward the partner to exceed 0.35; tail-to-tail requires both below -0.35; side-to-side has both absolute projections at most 0.5. All five displayed representatives were checked per cluster; the original clustering and PNGs were retained.

2.2 Parent selection and two-pH sequence redesign

Initial paired Boltz2 FAST assessments used 200 sampling steps, three recycles, one diffusion sample, and batch seed 0, with verified target MSAs and empty binder MSAs. The first parent selection retained ten from each species/epitope combination using mean human/mouse ipSAE multiplied by target-aligned human/mouse binder-pose TM similarity. A later top-epitope selection used mean ipSAE multiplied by mean RFD3-to-refolded binder-fold TM similarity. The labels P and RP identify these two parent-selection routes; they do not identify the source epitope by themselves. For example, P37 has a mouse top-epitope RFD3 ancestor.

Proton-PottsMPNN v6 used the epoch-0125 checkpoint. Conditional canonical decoder probabilities were averaged across the human and mouse parent backbones. Proposals were followed by 15 sweeps of two-pH Potts coordinate optimization. Temperatures were 0.05, 0.1, 0.2, and 0.4; each parent had a recorded seed. The pipeline built a pool of 160 accepted unique sequences to retain 100 per parent, subject to up to 2,000 attempts. Per-candidate proposal attempts, seeds, temperatures, mutable positions, exact substitutions, and objective values are reported below.

Up to 20 non-Cys binder positions nearest a target C-alpha within 10 angstrom in either species were mutable, ordered by summed distance. Target A/B/C, binder length, native Cys, and all other positions remained fixed, and new Cys were excluded. The generation-time interaction proxy was G = H(complex) - H(binder) - H(target trimer), with independent Henderson-Hasselbalch occupancies using intrinsic pKa H 6.5, D 3.9, and E 4.3. The minimized objective was 0.35 z(mean G7.4) + 0.40 z(mean G7.4 - mean G6.0) + 0.25 z(mean apo H7.4), with equal species weighting and frozen per-parent scales; z denotes division by the recorded scale, not recentering.

Acceptance required each species G7.4 to be no worse than its parent, G6.0 - G7.4 to be at least 0.10 model units and improve by at least 0.10 over the parent, apo H7.4 degradation no greater than 0.25 units per binder residue, and improvement of the composite objective. These are generation-time model guards. They neither calibrate affinity nor establish acidic release. Every retained exact descendant proceeded to paired human/mouse Boltz2 refolding. Optimization trajectories, baseline/scales, exported sequences and optimization records remain in each lineage record.

2.3 Expert substitutions and geometry-guided fusions

Expert variants apply only the named one-based substitutions to a selected Proton-PottsMPNN parent, leaving every other residue unchanged. They are deterministic sequence edits, followed by independent folding and scoring. They must not be described as an additional MPNN redesign. Candidate histories distinguish these expert edits from the preceding model-guided substitutions.

For fusions, exact two-domain sequences were concatenated in the geometry-selected order with a GGGGS-family linker. Parent-pair occupancy and receptor-overlay geometry preceded linker routing. A 2.0-angstrom grid with 3.0-angstrom atom-center exclusion sought solvent-accessible C-to-N paths. Linker length used 3.8 angstrom per residue at no more than 0.6 contour extension, at least five residues, a 3.0-angstrom separation between the three linker routes, and a 4.0-angstrom native-terminal launch zone. Candidate sections give the exact linker, domain boundaries, and maximum straight and routed distances. Parent overlays require receptor repacking and validation; favorable parent scores do not establish fusion performance.

The fusion generation record initially marked pH reassessment and refolding as pending. Subsequent Boltz2 Exact refolding and intact-fusion pH assessment supersede that generation-time status. Intact-fusion and extracted-domain observations remain distinct in the audit record. Three identical fusion chains D/E/F are modeled as separate copies; their two domains are covalently linked within each chain.

2.4 Current human/mouse models and pH assessment

Candidate panels use one common TNF-alpha trimer reference. Sequence-matched target C-alpha coordinates are fitted over all six protomer permutations using a proper rigid-body rotation; the same transform is applied to every atom of the complex. The top-epitope axis defines positive Z, and the side camera places positive Z upward. This aligns all 120 models across candidates, methods and species while preserving their native geometry and binding poses.

The current evidence comprises one independently confidence-selected model for each species and method: Boltz2 Exact, OpenDDE Exact, and standard AlphaFold3. Each request generated five samples at seed 0; maximum native confidence selected the displayed sample, with the lower sample index resolving ties. Selection did not optimize pH or ipSAE. Boltz2 Exact used 200 sampling steps and three recycles; OpenDDE Exact used 200 diffusion steps and ten cycles; AlphaFold3 used ten recycles. Native target sequences, disulfides, chain roles, input preparation, cached MSA/features, and method-specific software pins are retained in the source study. Standard AlphaFold3 here is separate from initial AFSample3 exploration.

Each model contains target A/B/C and three separate identical candidate copies D/E/F. Copy-specific ipSAE uses native PAE, a 10-angstrom cutoff, and best-protomer scoring; the weakest candidate copy is reported. The review threshold is 0.50. Cross-species agreement and each of three method pairs per species use symmetry-aware assembly DockQ, requiring at least 0.49 aggregate and 0.23 for the worst matched copy. These yield three cross-species method comparisons and six intermethod comparisons per candidate. Confidence and geometric agreement are computational diagnostics.

Post-refolding Proton-PottsMPNN assessment uses the selected native poses, selected binder His/Asp/Glu centers, and bound-complex PROPKA estimates. The same sequence positions are retained across species, methods, and copies, including translated domain positions in fusions. Each D/E/F copy is assessed against A/B/C separately, with receptor and noncenter ionization labels fixed. Independent-site expectations are evaluated at pH 6.0 and 7.4 under common pKa shifts -1, 0, and +1. The contrast S6.0 - S7.4 is positive when the proxy favors pH 7.4.

Potts interaction contrasts are in native model units; canonical teacher-forced MPNN contrasts are in nats per residue. Each model table reports the minimum over all copies and all pKa scenarios separately for both proxies. A robust-positive model requires both minima to be positive. The count out of six is evidence coverage across species/method combinations, not independent biological replication. Isolated-copy scoring does not estimate trivalent avidity, coupled protonation, or the full assembly binding equilibrium.

2.5 DeltaForge KD and selection scope

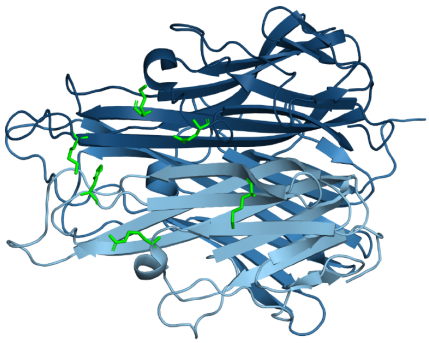
DeltaForge KD uses the same molecular stoichiometry for every candidate: one complete candidate monomer, chain D, against target trimer A/B/C, scored separately for human and mouse TNF-alpha. For a two-domain fusion, D contains both domains and the intact linker; E/F are omitted when extracting D from a three-copy source prediction. All 40 completed scoring receipts confirm four protein chains, receptor A/B/C, candidate D, the expected species-specific target lengths, and the full selected candidate length. The request and response input identities match, and the reported KD values agree with the selected-candidate record.

All 40 predictions use Boltz2-derived poses, SDK 0.16.4, the auto scorer request, and boltzmann_parallel aggregation. The returned scorer is v10_2_unified_parallel_2026_05_08 with Boltz2 calibration v10_boltz2_gbr_2026_03_29. KD is reported in native nM without a pH input. These are computational affinities of one candidate against a trimer, distinct from the three-copy structural panels and from a multimeric avidity measurement. Raw DeltaG is retained in its native units. The completed results are reused without rescoring.

The source selection retains four explicit manual overrides: P37-d002, F01, RP03-d017, and P37-d022. For other entries, at least one of the six selected models must have positive Potts and MPNN contrasts across all copies and pKa scenarios. Ordering prioritizes robust-positive model count, intermethod agreement count, cross-species agreement count, lower worse-species DeltaForge KD, ipSAE coverage, Boltz2 minimum-copy ipSAE, and candidate identity. We preserve the supplied CSV order and perform no new ranking. All entries remain research-review candidates; validation flags and manual inclusion are not altered by membership.

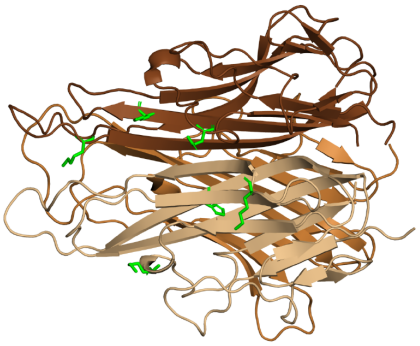
3 Side and top epitope context

A Human TNF-alpha - side epitope



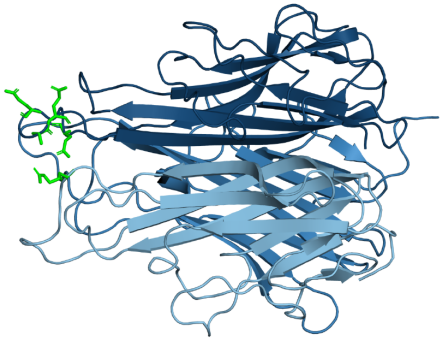
A: K65, K112, E146; C: H73, K90, R138

B Mouse TNF-alpha - side epitope



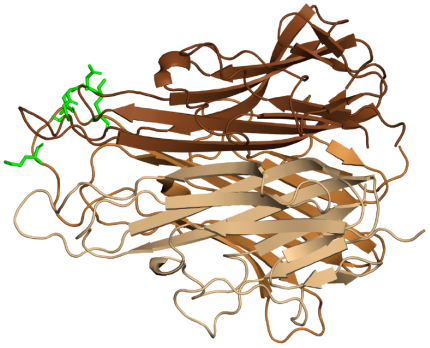
A: K65, K112, E146; C: H78, K90, K140

C Human TNF-alpha - top epitope



A: E104, E107, A109, E110; C: R103

D Mouse TNF-alpha - top epitope



A: D71, D104; B: E107, E110, K112

Green sticks: hotspot atoms | Target shades: protomers A/B/C | Source author residue numbering

Figure 1. Side and top epitopes on human and mouse TNF-alpha. A, human side epitope: A K65/K112/E146; C H73/K90/R138. B, mouse side epitope: A K65/K112/E146; C H78/K90/K140. C, human top epitope: A E104/E107/A109/E110; C R103. D, mouse top epitope: A D71/D104; B E107/E110/K112. Residues use source author numbering. Three blue shades identify human target protomers, three copper shades identify mouse target protomers, and green marks hotspot atoms. All four structural depictions retain their original common orientation and scale.

The retained structural panels depict the selected epitopes. The opening plates show the existing hexamer clusters and their interaction classes. A complete sample-by-sample causal map to the eventual hotspot selection is not recorded.

4 The selected 20

Table 1. Exact CSV order. pH is the number of robust-positive selected species/method models out of six; pose is passing intermethod pairs out of six. DeltaForge KD values score one complete candidate against human/mouse target trimer, in nM. An asterisk marks a manual override. Each candidate has one information page and one paired-view figure page.

Rank	Candidate	Sequence class	pH /6	Pose /6	KD H / M (nM)
1	P37-d002*	Proton-Potts redesign	6	6	42.48 / 43.05
2	RP06-d085-V60T	expert mutant	6	6	64.01 / 63.35
3	F07	GS fusion	6	1	21.19 / 15.17
4	F01*	GS fusion	6	1	37.71 / 13.22
5	RP03-d017*	Proton-Potts redesign	5	4	28.63 / 44.82
6	P18-d092	Proton-Potts redesign	5	4	48.28 / 33.26
7	P12-d084	Proton-Potts redesign	5	3	57.31 / 54.44
8	F02	GS fusion	5	1	15.85 / 15.41
9	P37-d022*	Proton-Potts redesign	4	6	50.84 / 44.36
10	RP06-d085	Proton-Potts redesign	4	4	42.21 / 46.40
11	F09	GS fusion	4	2	21.12 / 17.21
12	P02-d070	Proton-Potts redesign	4	1	49.88 / 59.55
13	F25	GS fusion	3	4	45.58 / 33.23
14	RP06-d085-V60S	expert mutant	3	2	65.39 / 65.44
15	F11	GS fusion	3	0	37.18 / 21.21
16	F32	GS fusion	2	6	37.37 / 23.37
17	RP03-d050-A40N_A69Q	expert mutant	2	4	62.17 / 73.52
18	RP03-d043-A40N_A69Q	expert mutant	2	4	79.22 / 74.35
19	P16-d085	Proton-Potts redesign	1	6	29.04 / 24.05
20	P00-d000	Proton-Potts redesign	1	1	17.26 / 13.07

The set comprises nine Proton-PottsMPNN redesigns, four subsequent expert mutants, and seven two-domain GS fusions. Candidate sections use one-based positions within each isolated parent sequence; fusion domain boundaries then map these positions onto the complete submitted chain. Only the 20 selected exact sequences are reported as candidates here; necessary ancestral domains are named as provenance intermediates.

5.01 P37-d002

Rank 1 | Proton-Potts redesign | 60 residues | Manual override retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P37-d002: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 2; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -412.893665. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3K, A12E, A16H, A39P, V40I, I42V, V45H, Q46H, T47A, Q48P, I53V (11 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 12.9270, G6.0-G7.4 4.7342; mouse G7.4 -25.1116, G6.0-G7.4 2.6896. G7.4 changes from parent: human -7.8463, mouse -4.0544. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 42.48 nM; mouse 43.05 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

EAKVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPIAVAFHHAPDGKLVAKVTFRV

5.01 P37-d002 - bound structures

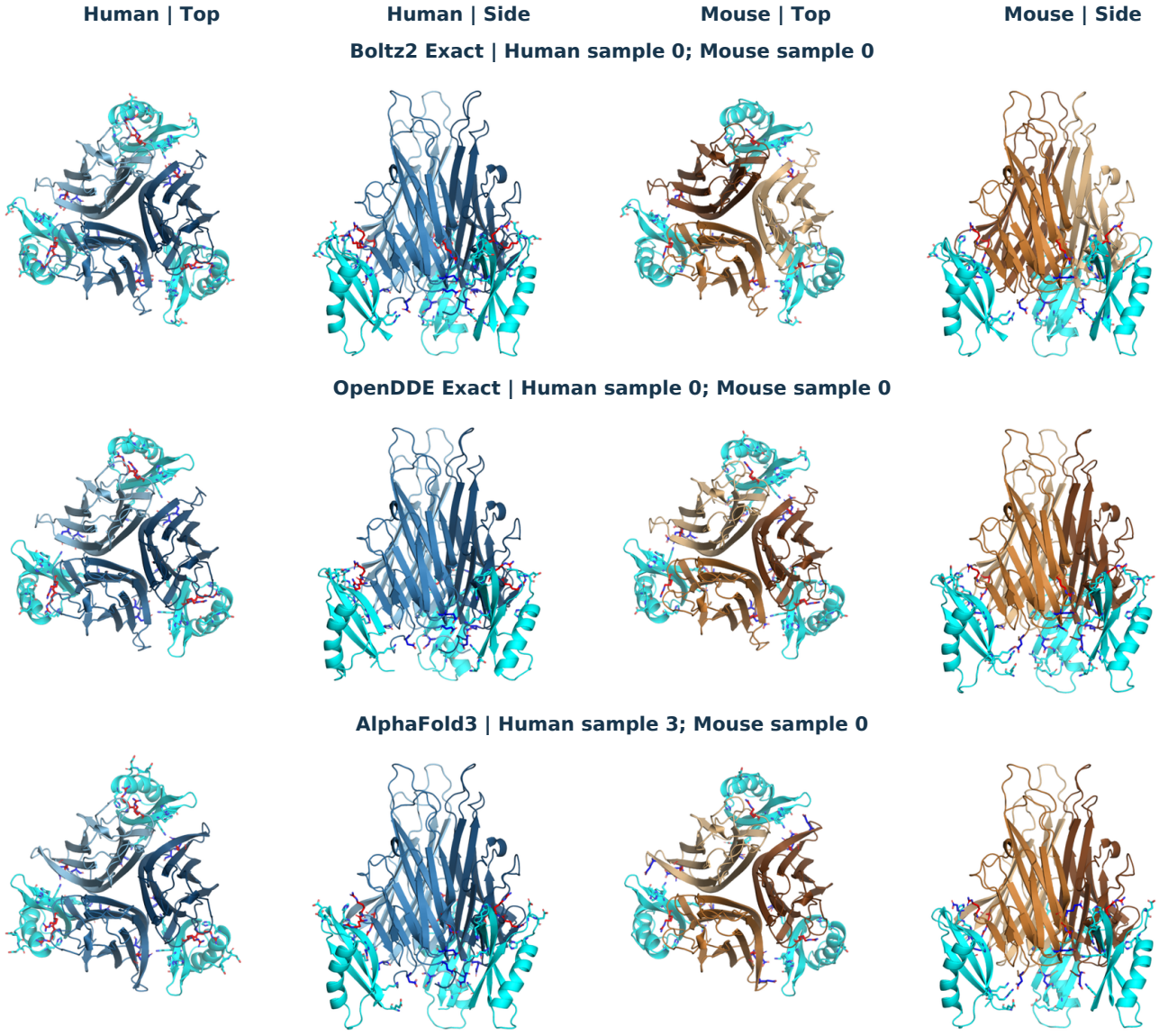


Figure 2. P37-d002: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 6/6; ipSAE pass: 2/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.846	0.1679	0.005992
Boltz2 / mouse / 0	0.834	0.02114	0.002118
OpenDDE / human / 0	0.040	0.9324	0.001971
OpenDDE / mouse / 0	0.210	0.105	0.003305
AF3 / human / 3	0.463	0.1033	0.004131
AF3 / mouse / 0	0.000	0.04628	0.004848

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 45.

5.02 RP06-d085-V60T

Rank 2 | expert mutant | 83 residues | Evidence-tier retained

Creation route. Foundry/RFD3 generation → Proton-PottsMPNN redesign of RP06-d085 → expert-specified V60T substitution(s).

RP06-d085: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 83-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 7.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20861028; accepted attempt 87; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -715.992329 to -719.792428. **Mutable positions (one-based):** 1, 36, 39, 40, 41, 62, 64, 66, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: S36H, V40R, S41Q, S62T, S64T, V66T, S69G, A71G, S73T, A74N, A75G, D76A, V78T (13 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 144.7273, G6.0-G7.4 3.6622; mouse G7.4 46.6635, G6.0-G7.4 1.5454. G7.4 changes from parent: human -10.7404, mouse -0.2608. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact expert mutation step

Expert-specified V60T substitutions were applied to RP06-d085. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP06-d085-V60T. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 64.01 nM; mouse 63.35 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

MAVTLQVAGNVTTVPAGATVTVTCAAAAAATTVTLHADGRQAAATTGSSASVSVTPTAGTHTYTCTATGA
GGTNGASTTVTVT

5.02 RP06-d085-V60T - bound structures

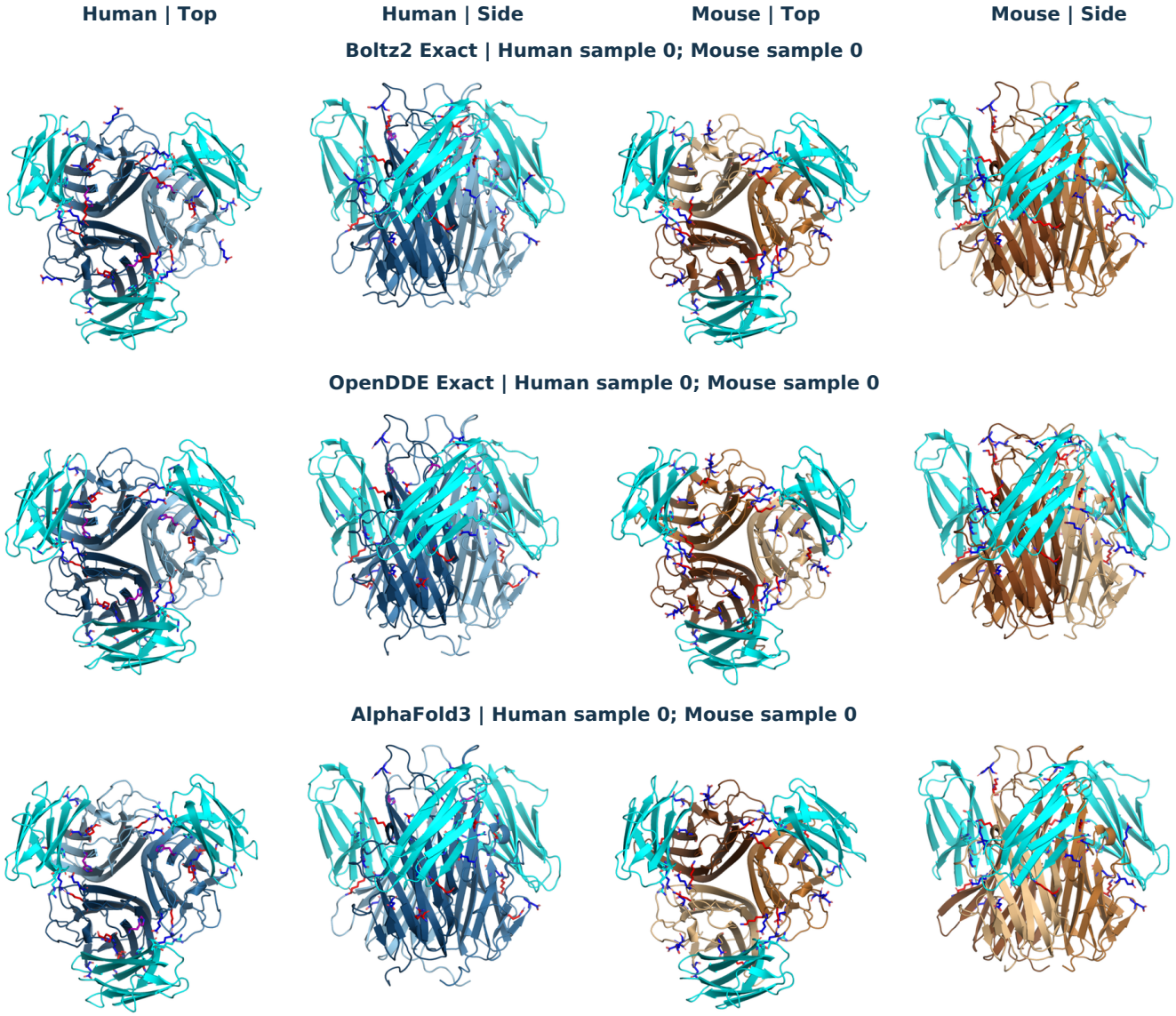


Figure 3. RP06-d085-V60T: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 6/6; ipSAE pass: 4/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.845	0.004502	1.939e-05
Boltz2 / mouse / 0	0.821	0.009326	5.188e-05
OpenDDE / human / 0	0.543	2.341	0.01703
OpenDDE / mouse / 0	0.577	0.004258	8.017e-05
AF3 / human / 0	0.000	0.002634	2.694e-05
AF3 / mouse / 0	0.000	0.008006	6.372e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 36, 38.

5.03 F07

Rank 3 | GS fusion | 163 residues | Evidence-tier retained

Creation route. P37-d022 → 22-residue GS linker → RP03-d050. Both independently redesigned domain histories and the exact fusion step follow.

Domain 1 - P37-d022: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 22; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -413.550655. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V (10 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 13.3211, G6.0-G7.4 5.7922; mouse G7.4 -23.4156, G6.0-G7.4 2.5233. G7.4 changes from parent: human -7.4522, mouse -2.3585. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d050: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 73; temperature 0.1; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -670.558797. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T21S, S28T, A49H, S51H, Q67T (5 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 54.8035, G6.0-G7.4 2.4292; mouse G7.4 115.1998, G6.0-G7.4 0.4890. G7.4 changes from parent: human -10.0744, mouse -4.9792. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d022 → linker → RP03-d050. The maximum C-to-N distance over the paired geometry was 47.98 angstrom and maximum solvent-route length 50.88 angstrom. Domain 1 occupies residues 1-60, the linker 61-82, and domain 2 83-163. Exact linker: GGGGSGGGGSGGGGSGGGGSGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 21.19 nM; mouse 15.17 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

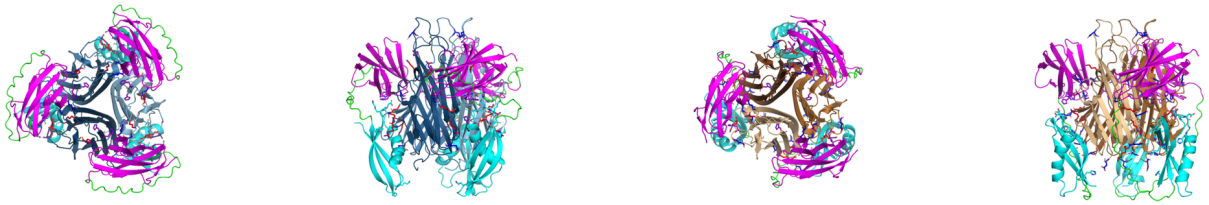
EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGSAAGPAVSAGASTTVTAGATLSVSASNATGAVTTTLSKDGATIAAAAATHAHFTVPATP
GTYTVTAATGADASSATVTVAAA

5.03 F07 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0

OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 0; Mouse sample 1

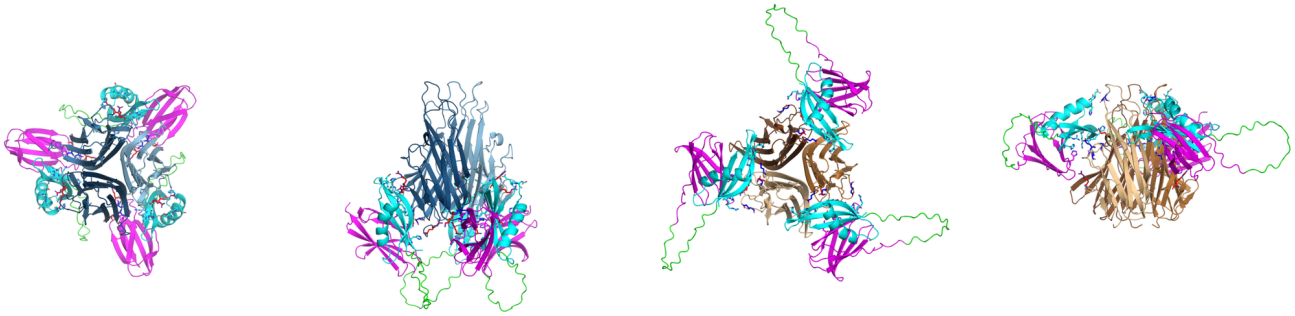


Figure 4. F07: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 6/6; ipSAE pass: 2/6; cross-species pose pass: 1/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.744	0.0881	0.0007851
Boltz2 / mouse / 0	0.744	0.1232	0.001698
OpenDDE / human / 0	0.074	1.012	0.0011
OpenDDE / mouse / 0	0.039	0.07099	0.001089
AF3 / human / 0	0.000	0.6119	0.003354
AF3 / mouse / 1	0.000	0.0172	0.0004073

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1, 133, 131.

5.04 F01

Rank 4 | GS fusion | 165 residues | Manual override retained

Creation route. P37-d002 → 24-residue GS linker → RP03-d017. AFSample3 animation: section 1.8.

Domain 1 - P37-d002: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 2; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -412.893665. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3K, A12E, A16H, A39P, V40I, I42V, V45H, Q46H, T47A, Q48P, I53V (11 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 12.9270, G6.0-G7.4 4.7342; mouse G7.4 -25.1116, G6.0-G7.4 2.6896. G7.4 changes from parent: human -7.8463, mouse -4.0544. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d017: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 23; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -672.193798. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N (8 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 53.9802, G6.0-G7.4 0.7972; mouse G7.4 108.5008, G6.0-G7.4 3.3428. G7.4 changes from parent: human -10.8977, mouse -11.6783. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d002 → linker → RP03-d017. The maximum C-to-N distance over the paired geometry was 52.65 angstrom and maximum solvent-route length 55.33 angstrom. Domain 1 occupies residues 1-60, the linker 61-84, and domain 2 85-165. Exact linker: GGGGSGGGGSGGGGSGGGGSGGGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 37.71 nM; mouse 13.22 nM. One complete candidate monomer against the respective TNF-alpha trimer.

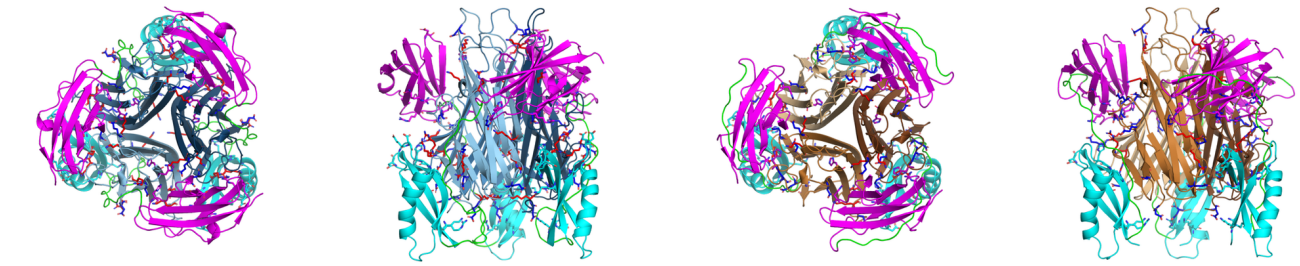
Exact selected sequence (N to C):

EAKVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPIAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGGGSAAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGATIAAAAAARATFTVPA
TPGTYTVTAANGADASSATVTVAAA

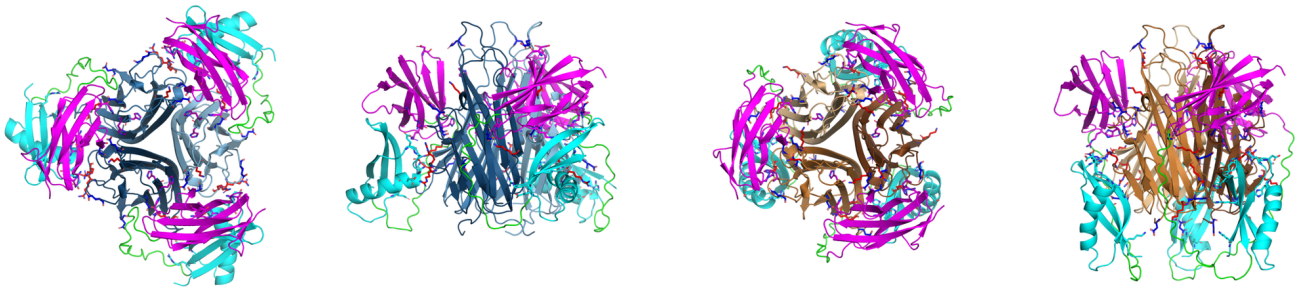
5.04 F01 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 1; Mouse sample 3

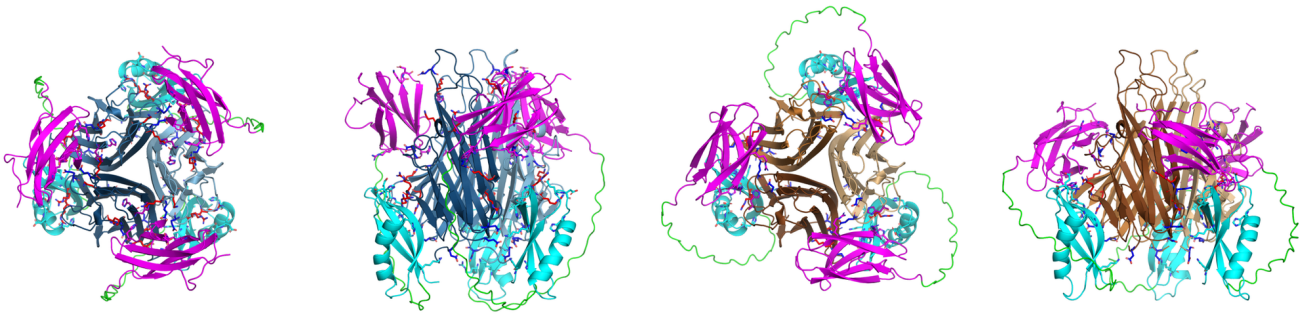


Figure 5. F01: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 6/6; ipSAE pass: 2/6; cross-species pose pass: 0/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.710	0.04959	0.0009204
Boltz2 / mouse / 0	0.771	0.03726	0.0004203
OpenDDE / human / 0	0.047	0.1587	0.006539
OpenDDE / mouse / 0	0.037	0.2312	0.0005135
AF3 / human / 1	0.000	0.2142	0.001249
AF3 / mouse / 3	0.000	0.001968	7.6e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 45, 114, 122.

5.05 RP03-d017

Rank 5 | Proton-Potts redesign | 81 residues | Manual override retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

RP03-d017: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 23; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -672.193798. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N (8 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 53.9802, G6.0-G7.4 0.7972; mouse G7.4 108.5008, G6.0-G7.4 3.3428. G7.4 changes from parent: human -10.8977, mouse -11.6783. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 28.63 nM; mouse 44.82 nM. One complete candidate monomer against the respective TNF-alpha trimer.

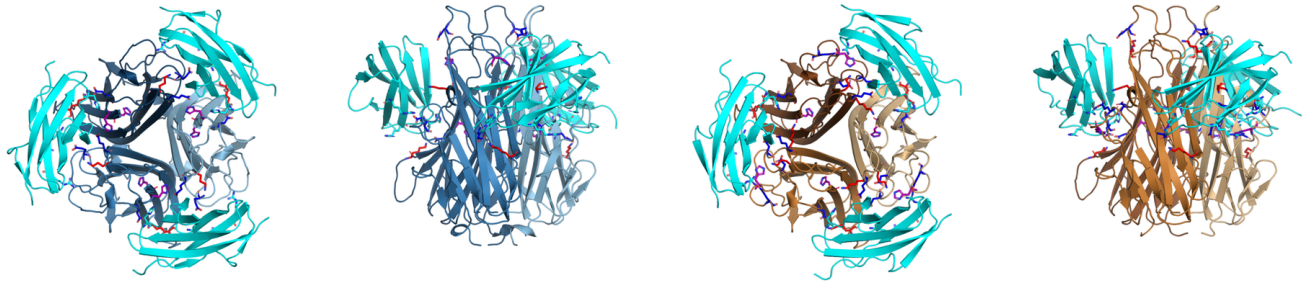
Exact selected sequence (N to C):

AAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGATIAAAAAARATFTVPATPGTYTVTAANGAD
ASSATVTVAAA

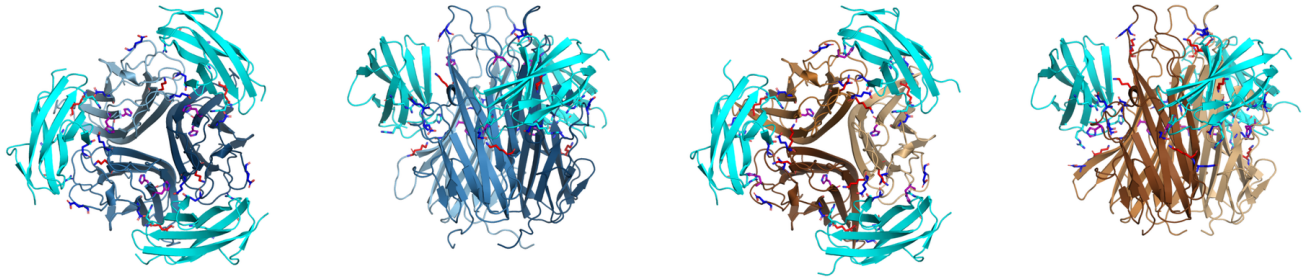
5.05 RP03-d017 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 1; Mouse sample 0

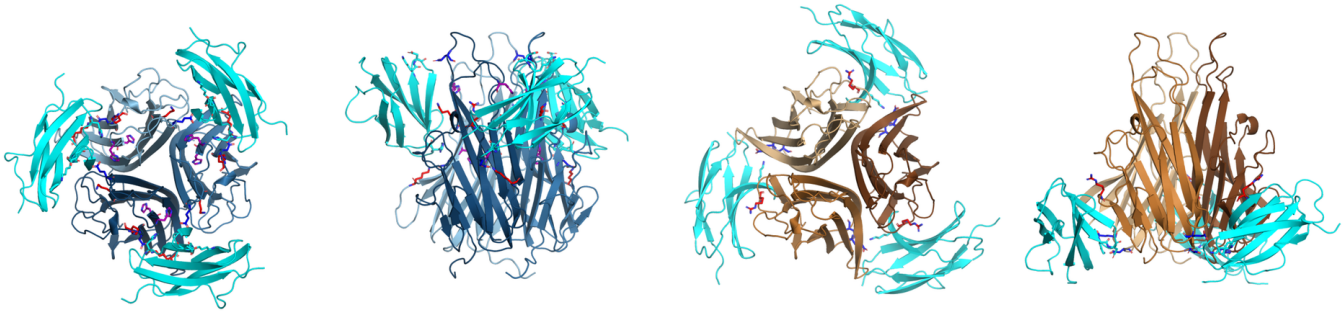


Figure 6. RP03-d017: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 5/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.824	0.002432	2.566e-05
Boltz2 / mouse / 0	0.843	0.002364	5.503e-05
OpenDDE / human / 0	0.449	0.01061	3.598e-05
OpenDDE / mouse / 0	0.387	0.01301	3.499e-05
AF3 / human / 1	0.202	0.003532	6.983e-05
AF3 / mouse / 0	0.000	-0.4338	-0.004906

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 30, 38.

5.06 P18-d092

Rank 6 | Proton-Potts redesign | 73 residues | Evidence-tier retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P18-d092: creation history

RFD3 generation. side epitope / mouse; complete trimer conditioned at A:LYS65, A:LYS112, A:GLU146, C:HIS78, C:LYS90, C:LYS140. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 73-residue parent. **Hotspot atom selectors:** A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C78: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C140: CB,CG,CD,CE,NZ.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 9.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 1800054; accepted attempt 94; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -575.859055 to -582.311822. **Mutable positions (one-based):** 20, 21, 22, 23, 24, 25, 26, 27, 28, 44, 45, 54, 55, 56, 57, 58, 59, 60, 62, 63. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: R23A, S24P, A27K, T28V, D44T, A45N, E55Q, T56H, F57Y, I58H, A60G, K62R (12 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 3.1359, G6.0-G7.4 3.0980; mouse G7.4 -9.1645, G6.0-G7.4 2.4289. G7.4 changes from parent: human -20.6686, mouse -17.7771. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 48.28 nM; mouse 33.26 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

MLSASELQAGDIVSLPSSVKAIAPGGKVIDLPAGSYKVIAVSSTNIALLDAATGQHYHVGSRVLNQNGGV
TLM

5.06 P18-d092 - bound structures

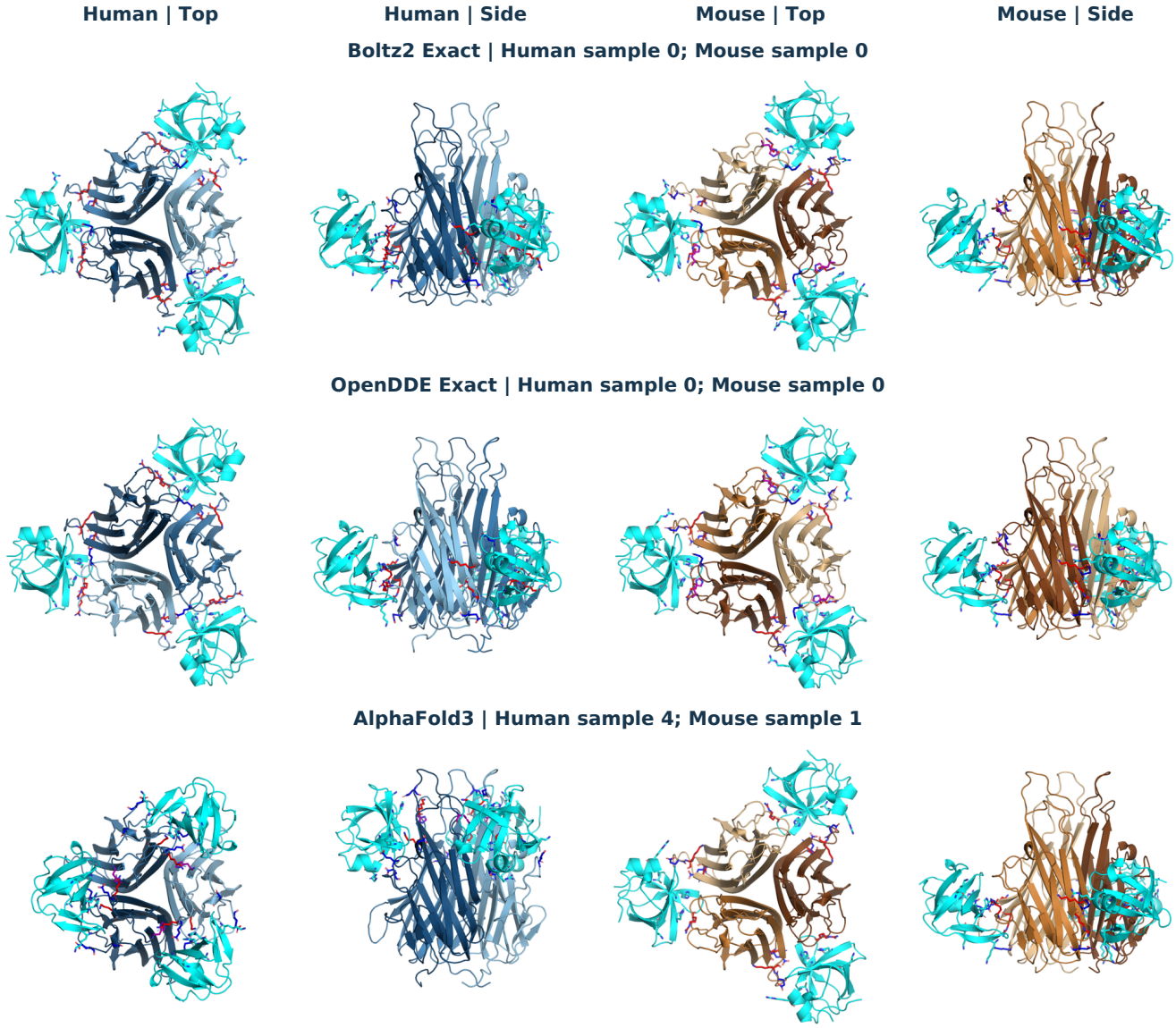


Figure 7. P18-d092: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 5/6; ipSAE pass: 4/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.708	0.008808	0.000252
Boltz2 / mouse / 0	0.651	0.0119	-2.193e-05
OpenDDE / human / 0	0.547	0.00437	5.904e-05
OpenDDE / mouse / 0	0.518	0.002171	4.89e-05
AF3 / human / 4	0.000	0.0112	0.0005285
AF3 / mouse / 1	0.464	0.002491	8.589e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 58, 56.

5.07 P12-d084

Rank 7 | Proton-Potts redesign | 75 residues | Evidence-tier retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P12-d084: creation history

RFD3 generation. side epitope / mouse; complete trimer conditioned at A:LYS65, A:LYS112, A:GLU146, C:HIS78, C:LYS90, C:LYS140. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 75-residue parent. **Hotspot atom selectors:** A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C78: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C140: CB,CG,CD,CE,NZ.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 3.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 1200036; accepted attempt 90; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -560.353410 to -563.850861. **Mutable positions (one-based):** 22, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 59, 60, 61, 62. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: Q22N, G35V, A36R, A42S, G43H, N48D, I49V, T59L, V61R, A62K (10 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 72.0461, G6.0-G7.4 0.6854; mouse G7.4 52.5755, G6.0-G7.4 1.0881. G7.4 changes from parent: human -13.9978, mouse -12.1394. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 57.31 nM; mouse 54.44 nM. One complete candidate monomer against the respective TNF-alpha trimer.

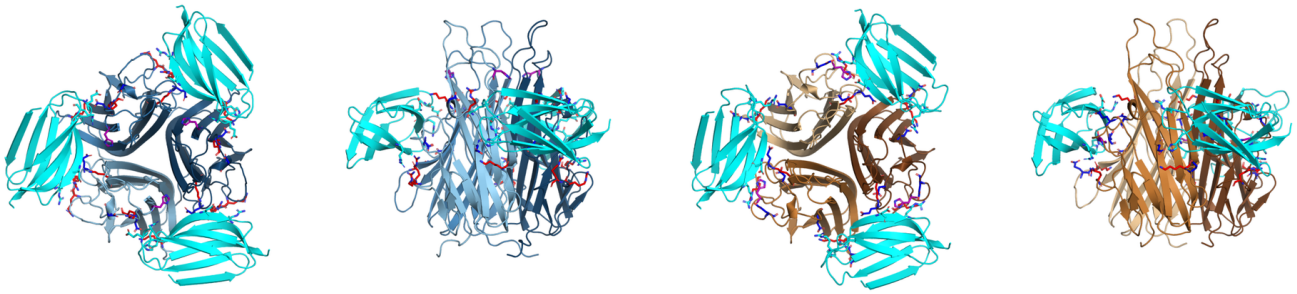
Exact selected sequence (N to C):

MKIEISVKNGIAQVAVTAVSENDVYTITLAGFNVVRVLAVSSHLTVLDVNAATGTLTYLARKGQDGVIVL
AVVPR

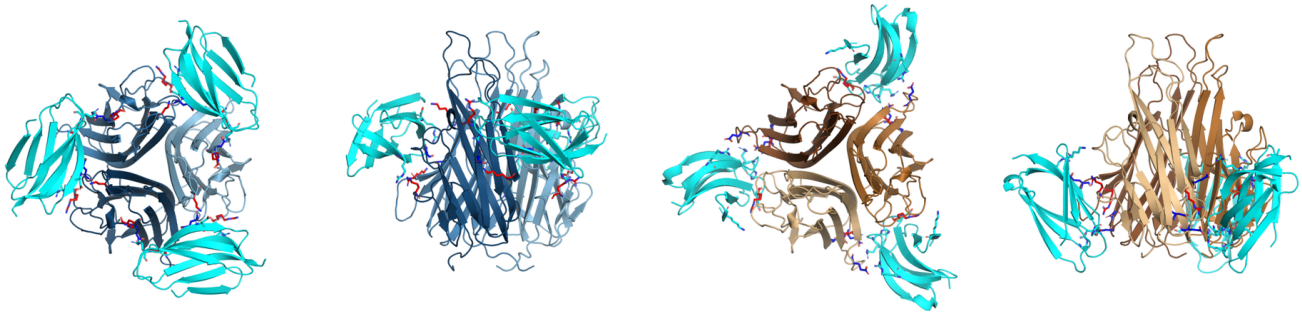
5.07 P12-d084 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 1; Mouse sample 2

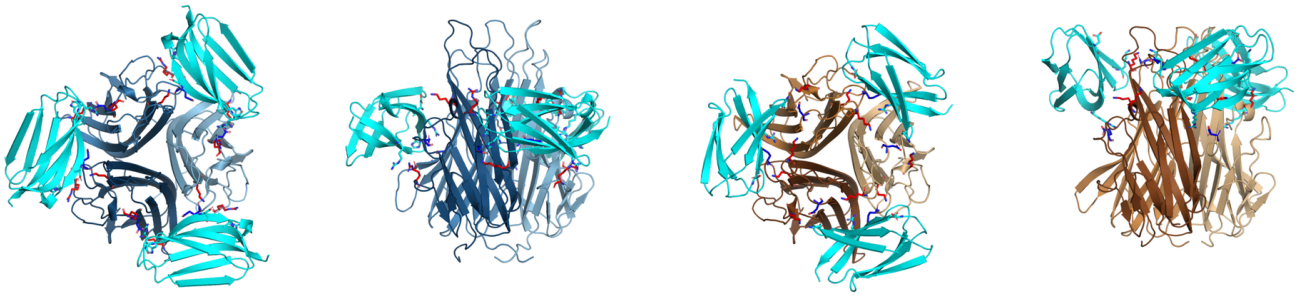


Figure 8. P12-d084: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 5/6; ipSAE pass: 3/6; cross-species pose pass: 1/3; intermethod pose pass: 3/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.832	-1.248	0.01024
Boltz2 / mouse / 0	0.809	0.007753	5.776e-05
OpenDDE / human / 0	0.308	0.02529	0.0005514
OpenDDE / mouse / 0	0.018	0.009171	0.0005305
AF3 / human / 1	0.526	0.02808	0.000416
AF3 / mouse / 2	0.000	0.006195	0.0004766

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 48, 21.

5.08 F02

Rank 8 | GS fusion | 163 residues | Evidence-tier retained

Creation route. P37-d002 → 22-residue GS linker → RP03-d050. Both independently redesigned domain histories and the exact fusion step follow.

Domain 1 - P37-d002: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 2; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -412.893665. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3K, A12E, A16H, A39P, V40I, I42V, V45H, Q46H, T47A, Q48P, I53V (11 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 12.9270, G6.0-G7.4 4.7342; mouse G7.4 -25.1116, G6.0-G7.4 2.6896. G7.4 changes from parent: human -7.8463, mouse -4.0544. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d050: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 73; temperature 0.1; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -670.558797. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T21S, S28T, A49H, S51H, Q67T (5 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 54.8035, G6.0-G7.4 2.4292; mouse G7.4 115.1998, G6.0-G7.4 0.4890. G7.4 changes from parent: human -10.0744, mouse -4.9792. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d002 → linker → RP03-d050. The maximum C-to-N distance over the paired geometry was 47.86 angstrom and maximum solvent-route length 50.30 angstrom. Domain 1 occupies residues 1-60, the linker 61-82, and domain 2 83-163. Exact linker: GGGGSGGGGSGGGGSGGGGSGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 15.85 nM; mouse 15.41 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

EAKVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPIAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGSAAGPAVSAGASTTVTAGATLSVSASNATGAVTTTLSKDGATIAAAAATHAHFTVPATP
GTYTVTAATGADASSATVTVAAA

5.08 F02 - bound structures

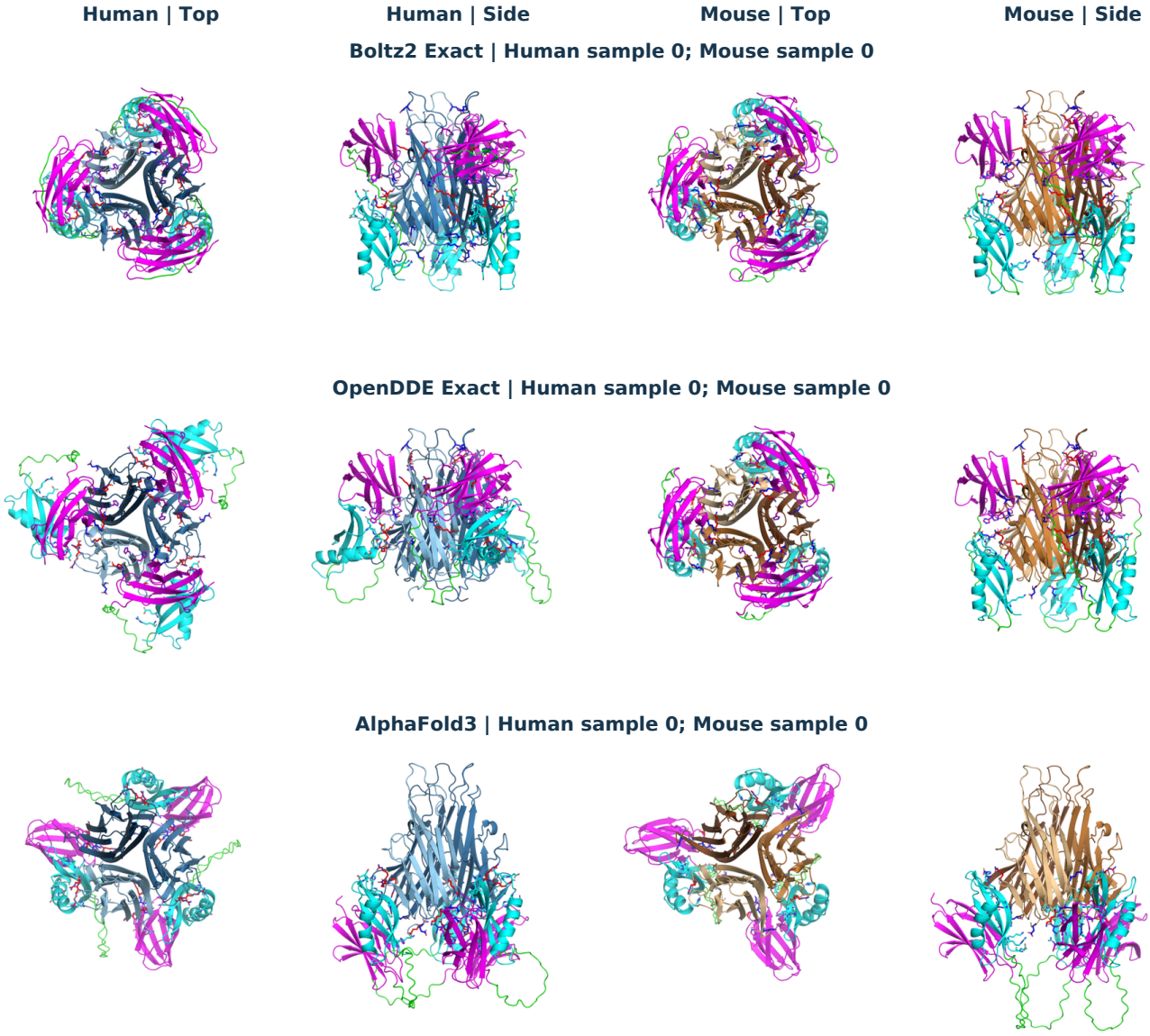


Figure 9. F02: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 5/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.662	0.04616	0.0011
Boltz2 / mouse / 0	0.720	0.7341	0.0125
OpenDDE / human / 0	0.067	0.3992	0.016
OpenDDE / mouse / 0	0.038	0.08544	0.002243
AF3 / human / 0	0.000	0.104	0.0004331
AF3 / mouse / 0	0.000	-0.6358	0.01128

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 45, 133, 131.

5.09 P37-d022

Rank 9 | Proton-Potts redesign | 60 residues | Manual override retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P37-d022: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 22; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -413.550655. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V (10 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 13.3211, G6.0-G7.4 5.7922; mouse G7.4 -23.4156, G6.0-G7.4 2.5233. G7.4 changes from parent: human -7.4522, mouse -2.3585. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 50.84 nM; mouse 44.36 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRV

5.09 P37-d022 - bound structures

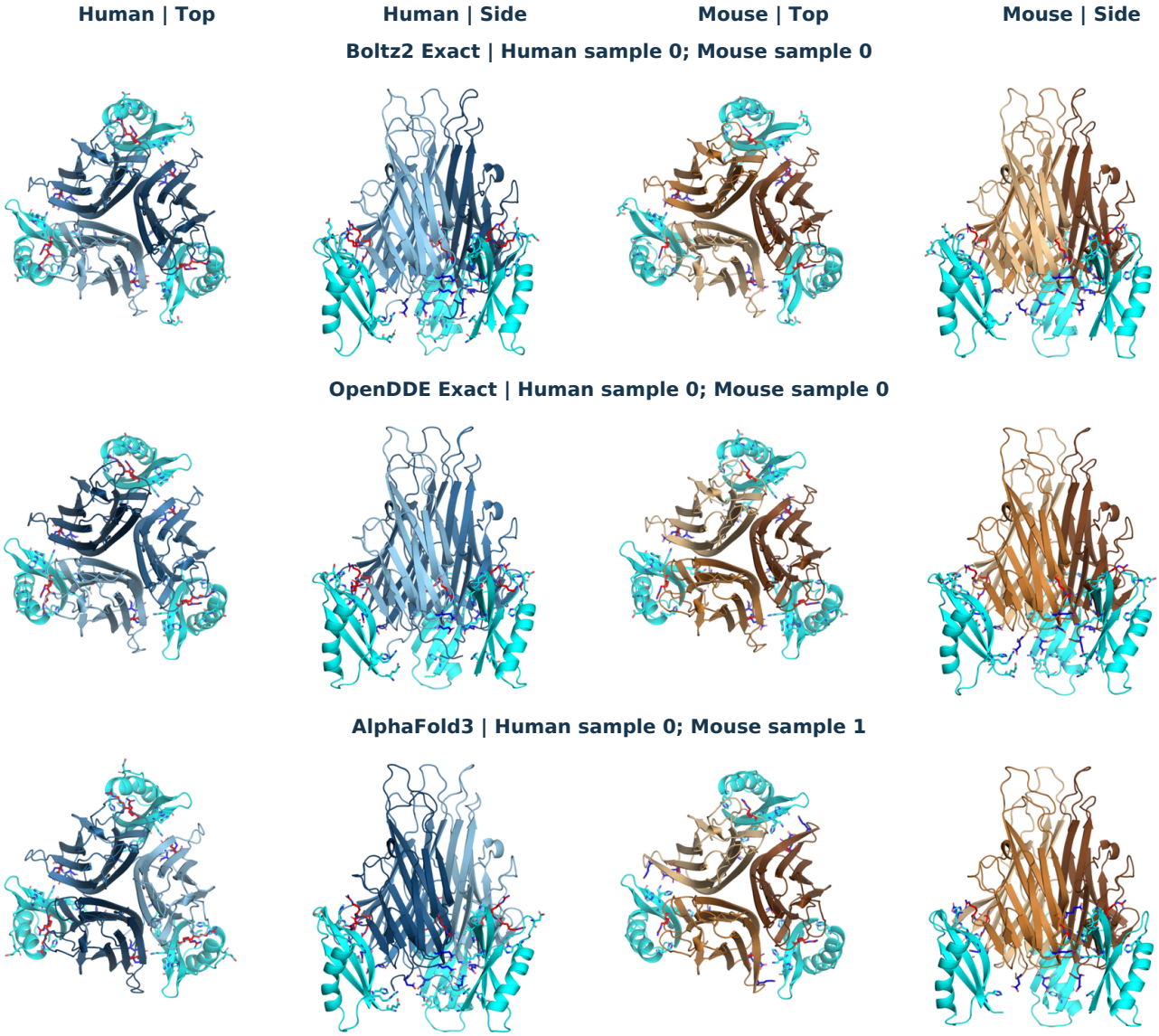


Figure 10. P37-d022: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 4/6; ipSAE pass: 2/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.878	0.05376	0.001811
Boltz2 / mouse / 0	0.769	0.0455	0.001771
OpenDDE / human / 0	0.178	0.8033	-0.01301
OpenDDE / mouse / 0	0.034	0.2684	0.008911
AF3 / human / 0	0.435	0.0928	0.003157
AF3 / mouse / 1	0.000	-0.118	0.008269

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1.

5.10 RP06-d085

Rank 10 | Proton-Potts redesign | 83 residues | Evidence-tier retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

RP06-d085: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 83-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 7.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20861028; accepted attempt 87; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -715.992329 to -719.792428. **Mutable positions (one-based):** 1, 36, 39, 40, 41, 62, 64, 66, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: S36H, V40R, S41Q, S62T, S64T, V66T, S69G, A71G, S73T, A74N, A75G, D76A, V78T (13 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 144.7273, G6.0-G7.4 3.6622; mouse G7.4 46.6635, G6.0-G7.4 1.5454. G7.4 changes from parent: human -10.7404, mouse -0.2608. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 42.21 nM; mouse 46.40 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

MAVTLQVAGNVTTVPAGATVTVTCAAAAAATTVTLHADGRQAAATTGSSASVSVTPTAGVHTYTCTATGA
GGTNGASTTVTVT

5.10 RP06-d085 - bound structures

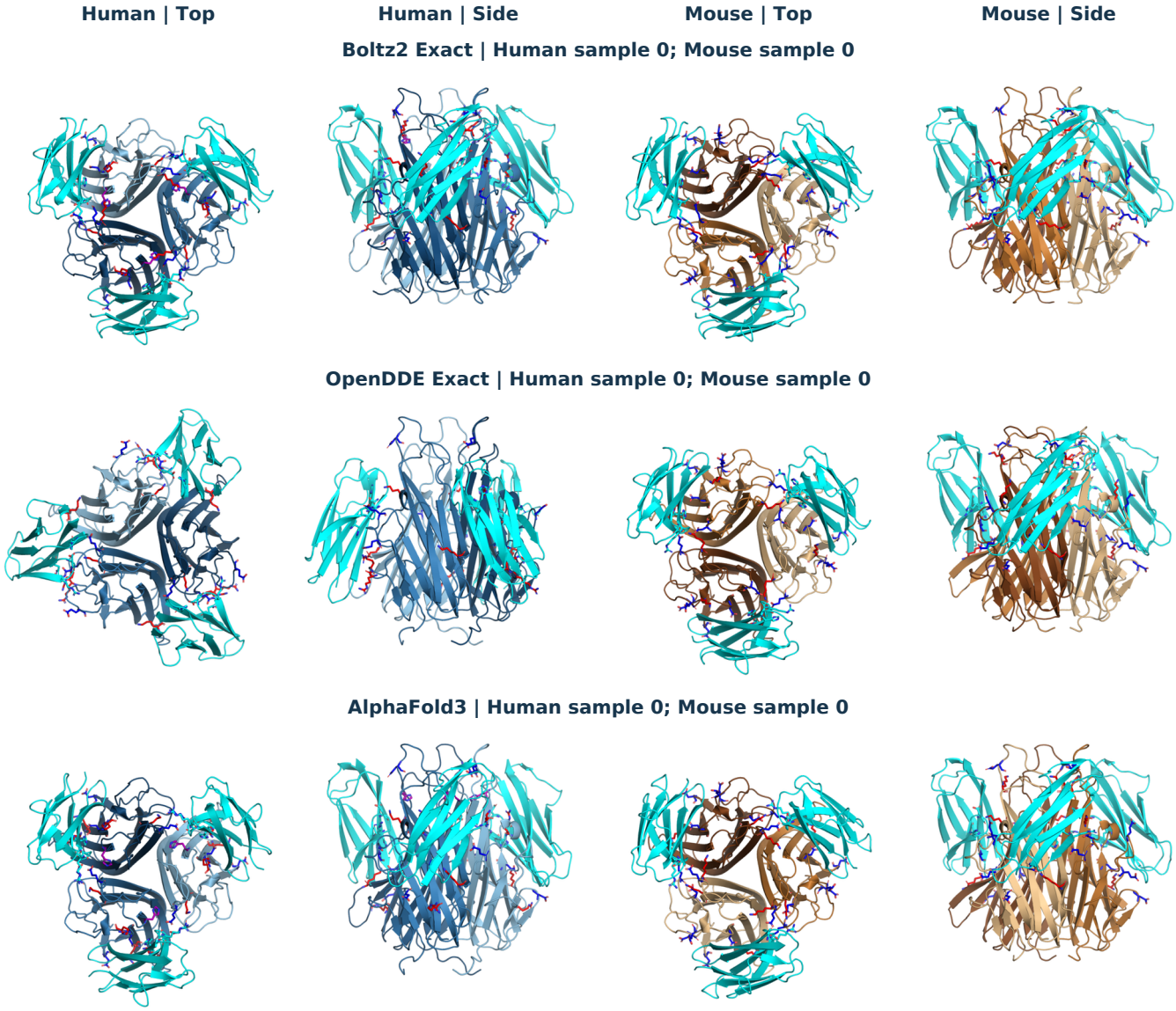


Figure 11. RP06-d085: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 4/6; ipSAE pass: 4/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.825	0.004672	2.269e-05
Boltz2 / mouse / 0	0.855	0.01145	0.0001187
OpenDDE / human / 0	0.016	0.05624	-0.005319
OpenDDE / mouse / 0	0.585	0.006379	1.988e-05
AF3 / human / 0	0.619	-0.0006657	-0.0005708
AF3 / mouse / 0	0.000	0.00963	7e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 36, 38.

5.11 F09

Rank 11 | GS fusion | 164 residues | Evidence-tier retained

Creation route. P37-d022 → 23-residue GS linker → RP03-d043. Both independently redesigned domain histories and the exact fusion step follow.

Domain 1 - P37-d022: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 22; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -413.550655. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V (10 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 13.3211, G6.0-G7.4 5.7922; mouse G7.4 -23.4156, G6.0-G7.4 2.5233. G7.4 changes from parent: human -7.4522, mouse -2.3585. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d043: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 64; temperature 0.05; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -669.254062. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, A27H, T32L, T48S, A49R, S51T (6 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 57.6741, G6.0-G7.4 0.8766; mouse G7.4 104.6333, G6.0-G7.4 0.1625. G7.4 changes from parent: human -7.2038, mouse -15.5458. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d022 → linker → RP03-d043. The maximum C-to-N distance over the paired geometry was 50.61 angstrom and maximum solvent-route length 53.08 angstrom. Domain 1 occupies residues 1-60, the linker 61-83, and domain 2 84-164. Exact linker: GGGGSGGGGSGGGGSGGGGSGGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 21.12 nM; mouse 17.21 nM. One complete candidate monomer against the respective TNF-alpha trimer.

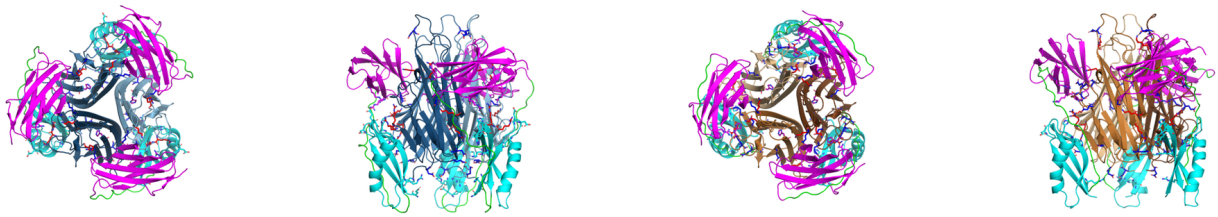
Exact selected sequence (N to C):

EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGGSAAGPAVSAGASTTVTAGARLTVSASNHSGAVLTTLSKDGATIAAAAASRATFTVPAT
PGTYTVTAAQGADASSATVTVAAA

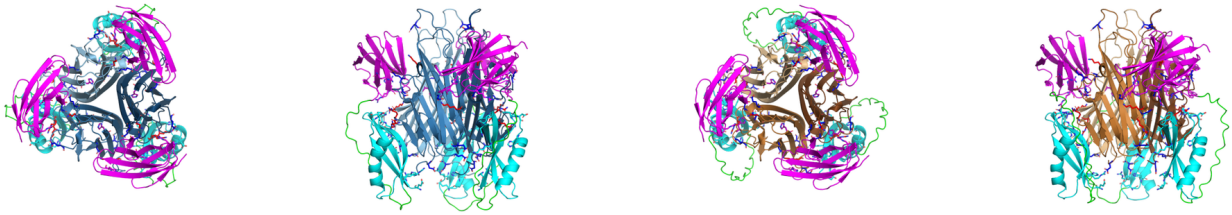
5.11 F09 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 0; Mouse sample 3

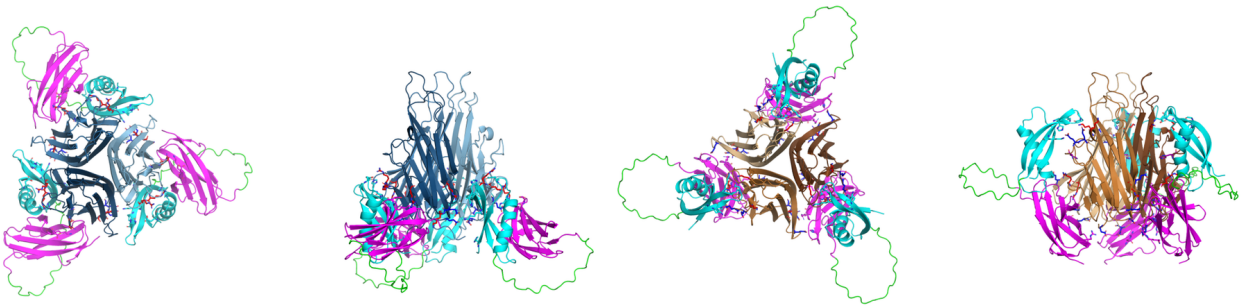


Figure 12. F09: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 4/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 2/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.764	0.04201	0.0002242
Boltz2 / mouse / 0	0.757	0.04487	0.001143
OpenDDE / human / 0	0.033	0.9914	0.002518
OpenDDE / mouse / 0	0.043	0.7582	-0.01249
AF3 / human / 0	0.000	0.04124	0.005063
AF3 / mouse / 3	0.000	-0.02139	-0.00675

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1, 110, 153.

5.12 P02-d070

Rank 12 | Proton-Potts redesign | 73 residues | Evidence-tier retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P02-d070: creation history

RFD3 generation. side epitope / human; complete trimer conditioned at A:LYS65, A:LYS112, A:GLU146, C:HIS73, C:LYS90, C:ARG138. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 73-residue parent. **Hotspot atom selectors:** A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C73: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C138: CB,CG,CD,NE,CZ,NH1,NH2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 3.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 200006; accepted attempt 74; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -550.317242 to -558.700326. **Mutable positions (one-based):** 15, 16, 17, 19, 24, 25, 26, 27, 31, 32, 33, 58, 60, 61, 62, 63, 64, 65, 66, 67. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: Q19S, N25Q, A31P, L33R, N58H, I60W, K63R, D65H, K66H, V67R (10 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 -29.0966, G6.0-G7.4 4.1500; mouse G7.4 8.3285, G6.0-G7.4 5.4732. G7.4 changes from parent: human -6.4302, mouse -1.2649. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 49.88 nM; mouse 59.55 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

AATSVSIALTAPVGAYLVSAYVGGQLVAAEPLRTDAAGDVSKTLSVAALASGDAVSLHVWDVRSHHRVAS
EAV

5.12 P02-d070 - bound structures

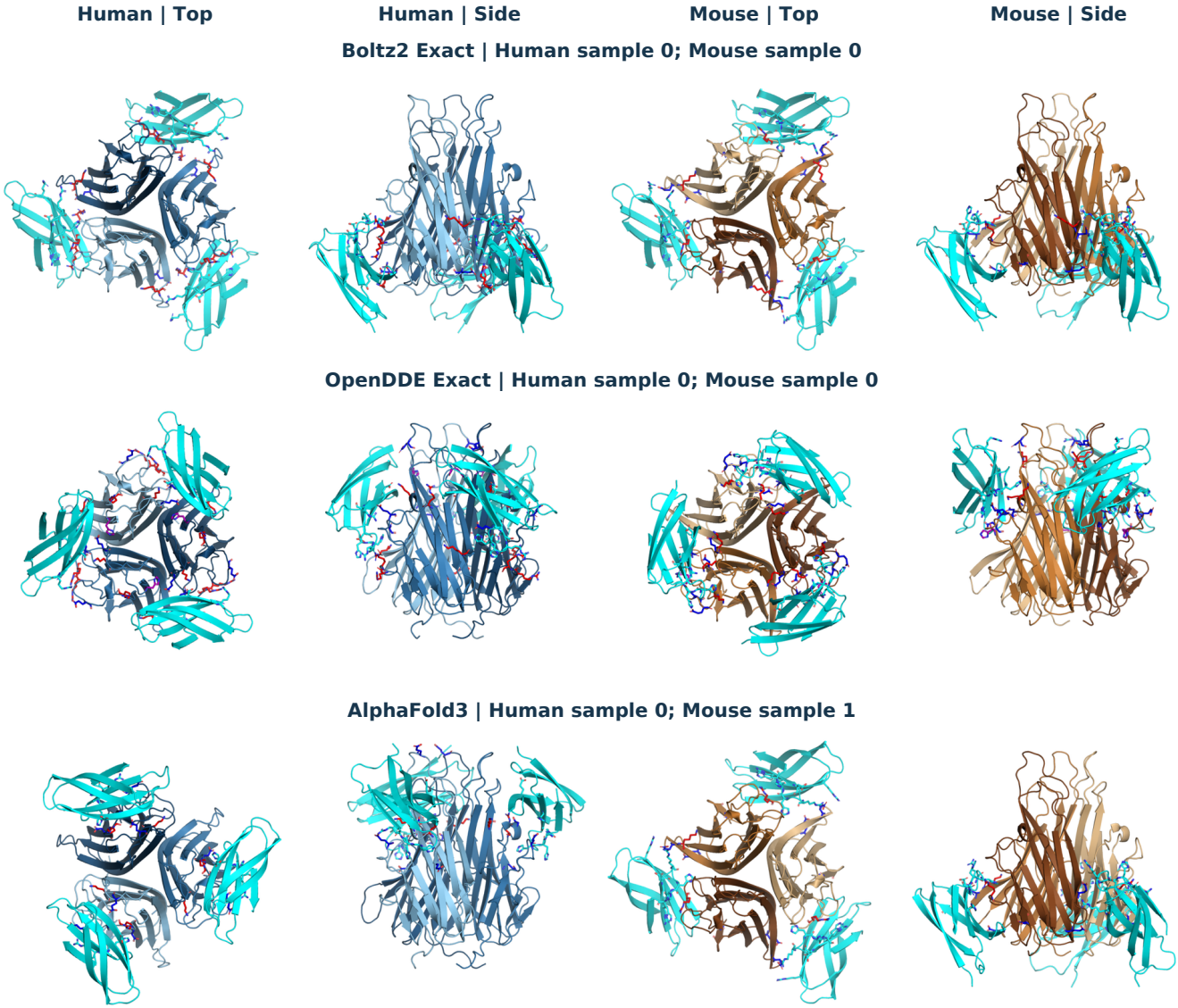


Figure 13. P02-d070: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 4/6; ipSAE pass: 3/6; cross-species pose pass: 1/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.693	0.1222	0.004672
Boltz2 / mouse / 0	0.687	0.1861	0.005234
OpenDDE / human / 0	0.018	0.1547	0.002648
OpenDDE / mouse / 0	0.514	-1.598	0.002472
AF3 / human / 0	0.000	-0.03634	0.004125
AF3 / mouse / 1	0.000	0.2166	0.00387

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 65, 66.

5.13 F25

Rank 13 | GS fusion | 180 residues | Evidence-tier retained

Creation route. RP11-d064 → 19-residue GS linker → RP03-d017. Both independently redesigned domain histories and the exact fusion step follow.

Domain 1 - RP11-d064: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 80-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 2.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 21361043; accepted attempt 69; temperature 0.1; 15 two-pH optimization sweeps (section 2.2). Composite objective: -655.969112 to -663.136294. **Mutable positions (one-based):** 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 54, 55, 56, 57, 58, 59, 64. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: L35M, T36V, A40G, I41V, A42D, S43H, G56K, S57H, S59A (9 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 19.0508, G6.0-G7.4 3.8665; mouse G7.4 -13.6937, G6.0-G7.4 4.5458. G7.4 changes from parent: human -10.4492, mouse -17.1299. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d017: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 23; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -672.193798. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N (8 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 53.9802, G6.0-G7.4 0.7972; mouse G7.4 108.5008, G6.0-G7.4 3.3428. G7.4 changes from parent: human -10.8977, mouse -11.6783. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified RP11-d064 → linker → RP03-d017. The maximum C-to-N distance over the paired geometry was 42.05 angstrom and maximum solvent-route length 43.50 angstrom. Domain 1 occupies residues 1-80, the linker 81-99, and domain 2 100-180. Exact linker: GGGGSGGGGSGGGGSGGGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 45.58 nM; mouse 33.23 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

VSTATLPAPGDSTTTPPAASATVTVLSGKVSLTAMVGGAGVDHATLTPGQSHAFPKHAAQMTLVATNVGT
ATATIEYNTTGGGGSGGGGSGGGGSGGGSAAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGAT
IAAAAAARATFTVPATPGTYTVTAANGADASSATVTVAAA

5.13 F25 - bound structures

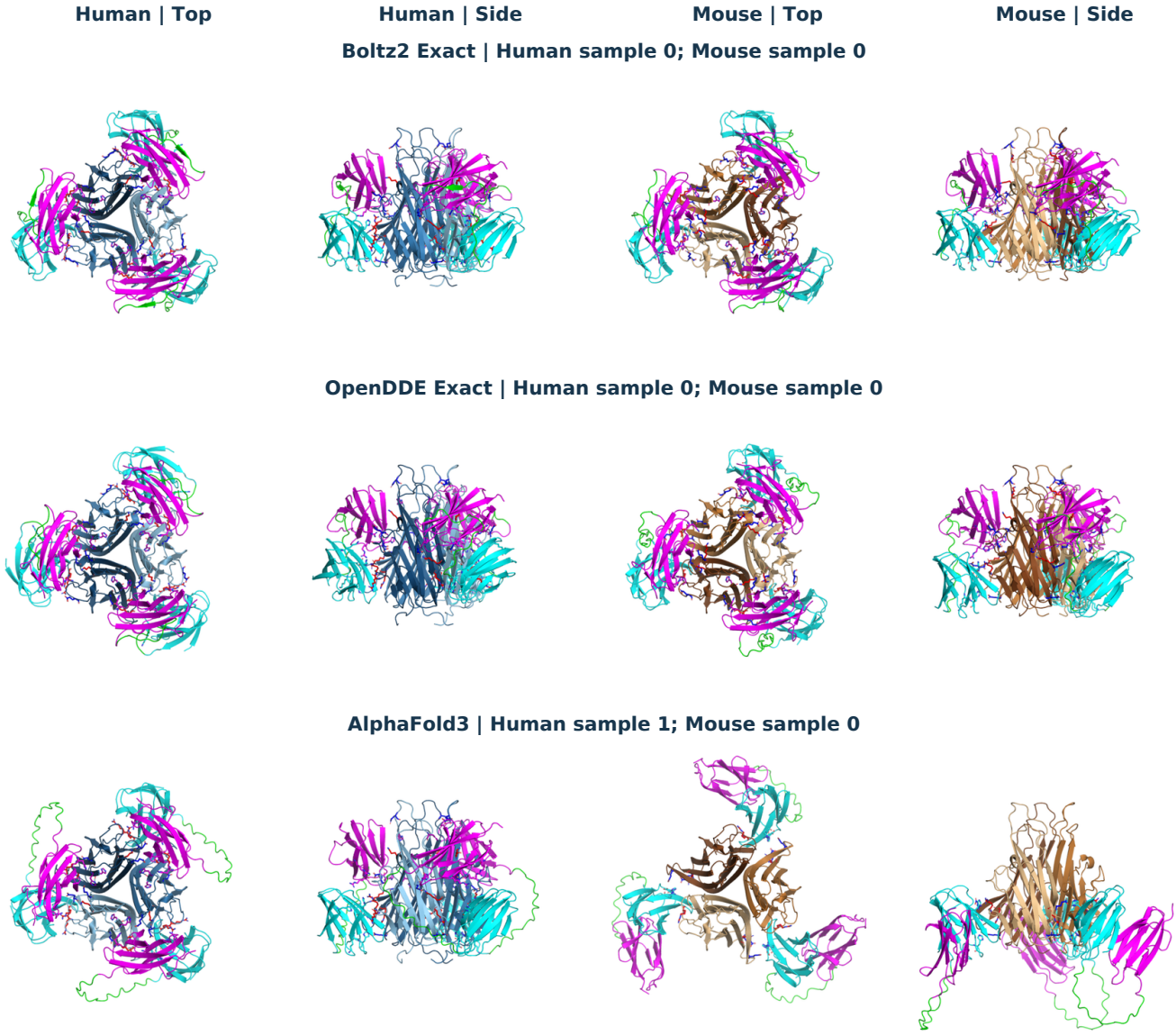


Figure 14. F25: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 3/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.654	-0.7163	-0.01296
Boltz2 / mouse / 0	0.795	-0.6824	-0.01926
OpenDDE / human / 0	0.036	0.008617	1.499e-05
OpenDDE / mouse / 0	0.047	0.01152	0.000178
AF3 / human / 1	0.000	0.4393	0.003987
AF3 / mouse / 0	0.000	0.2479	-0.01544

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 57, 43, 129, 137.

5.14 RP06-d085-V60S

Rank 14 | expert mutant | 83 residues | Evidence-tier retained

Creation route. Foundry/RFD3 generation → Proton-PottsMPNN redesign of RP06-d085 → expert-specified V60S substitution(s).

RP06-d085: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 83-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 7.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20861028; accepted attempt 87; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -715.992329 to -719.792428. **Mutable positions (one-based):** 1, 36, 39, 40, 41, 62, 64, 66, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: S36H, V40R, S41Q, S62T, S64T, V66T, S69G, A71G, S73T, A74N, A75G, D76A, V78T (13 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 144.7273, G6.0-G7.4 3.6622; mouse G7.4 46.6635, G6.0-G7.4 1.5454. G7.4 changes from parent: human -10.7404, mouse -0.2608. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact expert mutation step

Expert-specified V60S substitutions were applied to RP06-d085. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP06-d085-V60S. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 65.39 nM; mouse 65.44 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

MAVTLQVAGNVTTVPAGATVTVTCAAAAAATTVTLHADGRQAAATTGSSASVSVTPTAGSHTYTCTATGA
GGTNGASTTVTVT

5.14 RP06-d085-V60S - bound structures

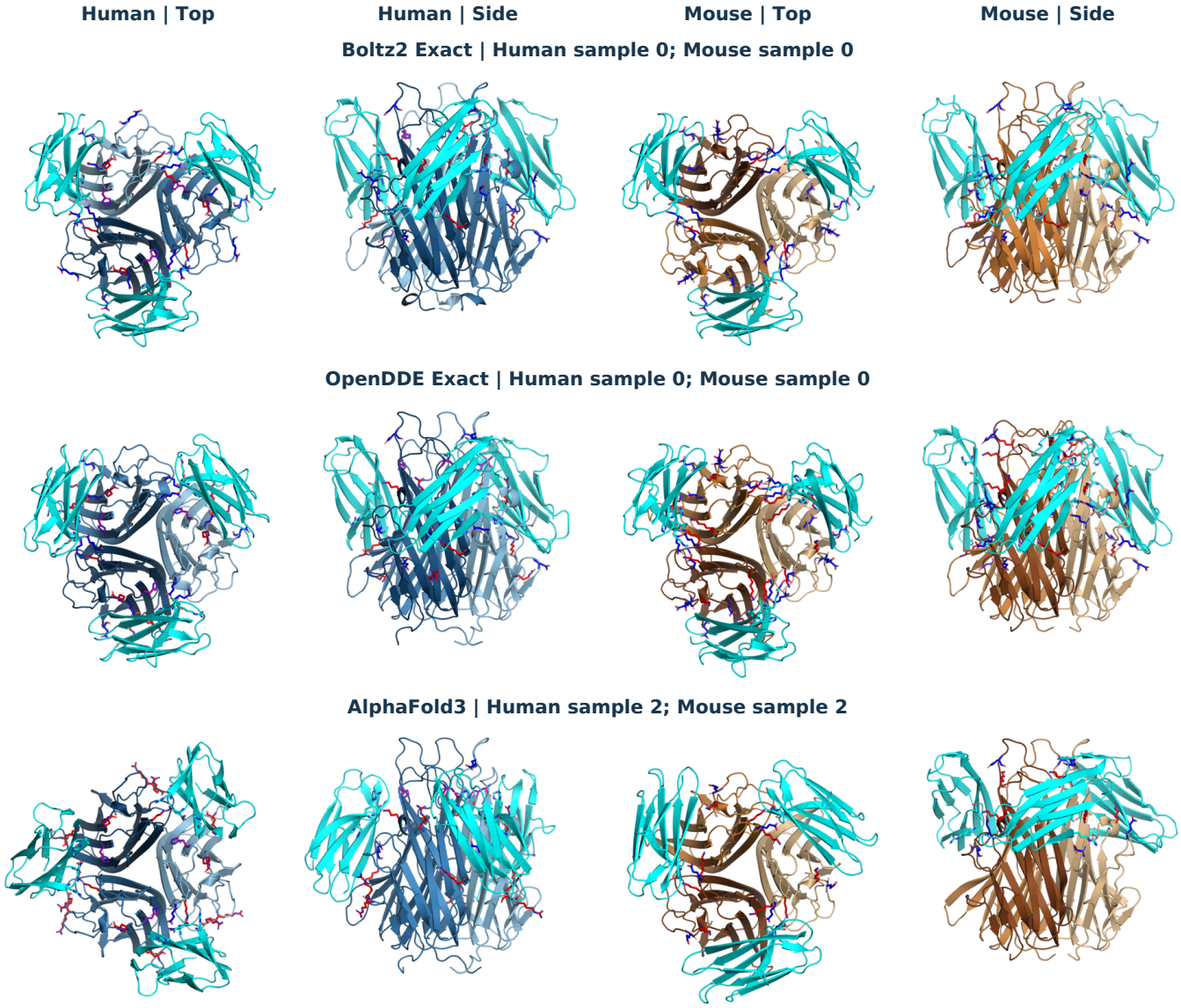


Figure 15. RP06-d085-V60S: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 3/6; ipSAE pass: 3/6; cross-species pose pass: 2/3; intermethod pose pass: 2/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.823	-0.07061	0.0006047
Boltz2 / mouse / 0	0.852	0.01056	-0.008794
OpenDDE / human / 0	0.496	0.00449	1.317e-05
OpenDDE / mouse / 0	0.656	0.006253	-0.006721
AF3 / human / 2	0.000	0.01755	0.0001598
AF3 / mouse / 2	0.000	0.01311	1.959e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 36, 38.

5.15 F11

Rank 15 | GS fusion | 163 residues | Evidence-tier retained

Creation route. P37-d022 → 22-residue GS linker → RP03-d081. Both independently redesigned domain histories and the exact fusion step follow.

Domain 1 - P37-d022: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 60-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 8.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3700111; accepted attempt 22; temperature 0.2; 15 two-pH optimization sweeps (section 2.2). Composite objective: -405.867972 to -413.550655. **Mutable positions (one-based):** 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V (10 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 13.3211, G6.0-G7.4 5.7922; mouse G7.4 -23.4156, G6.0-G7.4 2.5233. G7.4 changes from parent: human -7.4522, mouse -2.3585. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d081: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 111; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -671.910564. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, T21R, S28T, A49H, S51V, Q67S (6 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 47.7405, G6.0-G7.4 3.4747; mouse G7.4 112.0606, G6.0-G7.4 0.1652. G7.4 changes from parent: human -17.1374, mouse -8.1185. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d022 → linker → RP03-d081. The maximum C-to-N distance over the paired geometry was 48.98 angstrom and maximum solvent-route length 52.12 angstrom. Domain 1 occupies residues 1-60, the linker 61-82, and domain 2 83-163. Exact linker: GGGGSGGGGSGGGGSGGGGSGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 37.18 nM; mouse 21.21 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGSAAGPAVSAGASTTVTAGARLRVSASNATGAVTTTLSKDGATIAAAAATHAVFTVPATP
GTYTVTAASGADASSATVTVAAA

5.15 F11 - bound structures

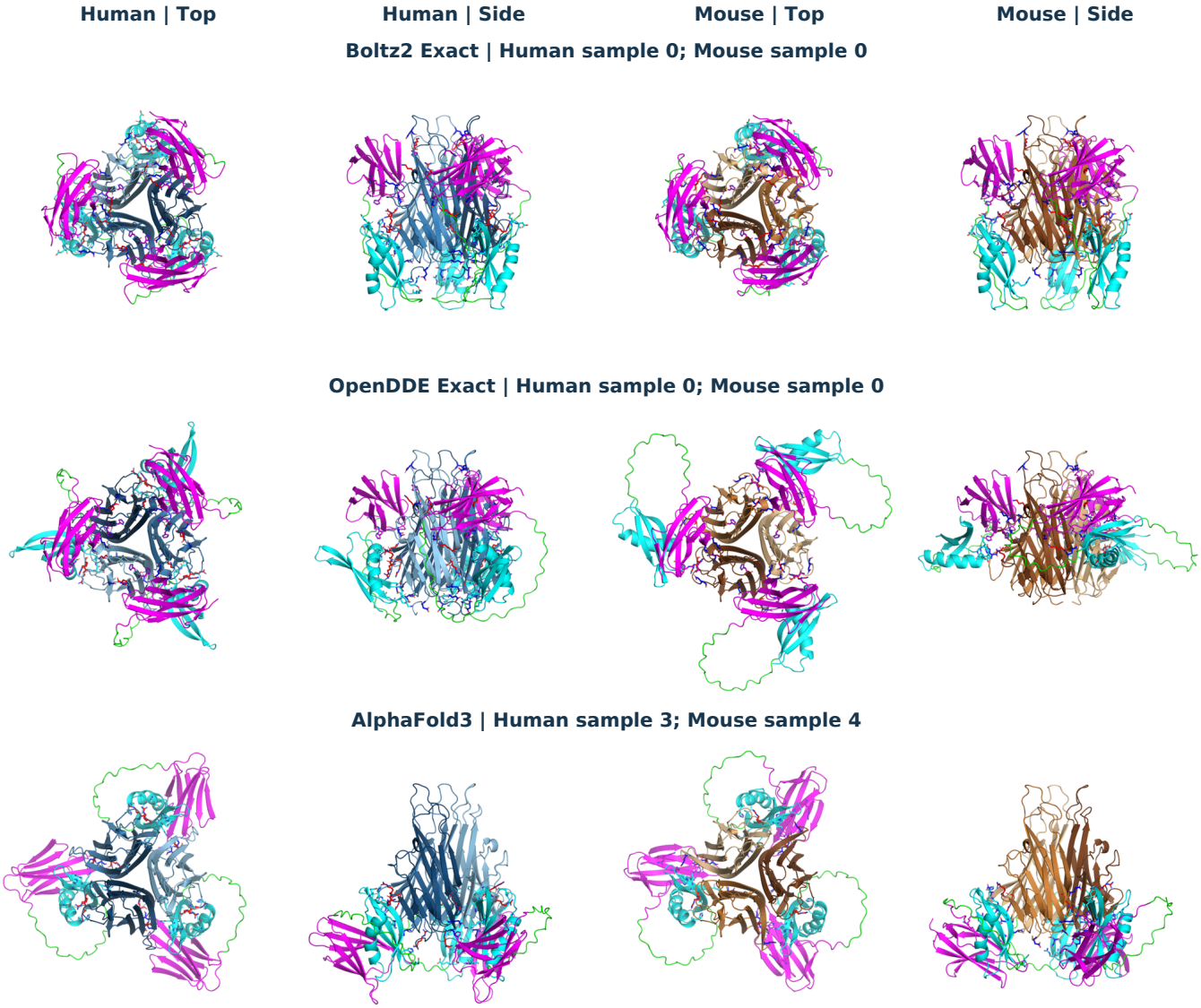


Figure 16. F11: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 3/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 0/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.753	0.03785	0.0001427
Boltz2 / mouse / 0	0.710	0.03676	0.0005054
OpenDDE / human / 0	0.052	0.1366	0.001704
OpenDDE / mouse / 0	0.012	0.8863	-0.02054
AF3 / human / 3	0.128	0.5519	-0.008957
AF3 / mouse / 4	0.000	-3.231	0.001497

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1, 131, 120.

5.16 F32

Rank 16 | GS fusion | 179 residues | Evidence-tier retained

Creation route. P30-d032 → 18-residue GS linker → RP03-d017. Both independently redesigned domain histories and the exact fusion step follow.

Domain 1 - P30-d032: creation history

RFD3 generation. top epitope / mouse; complete trimer conditioned at B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 80-residue parent. **Hotspot atom selectors:** B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 1.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 3000090; accepted attempt 33; temperature 0.1; 15 two-pH optimization sweeps (section 2.2). Composite objective: -655.969112 to -663.172171. **Mutable positions (one-based):** 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 54, 55, 56, 57, 58, 59, 64. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: L35M, T36V, A40G, I41V, A42D, S43H, G56P, S57H (8 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 16.4747, G6.0-G7.4 3.8044; mouse G7.4 -12.5324, G6.0-G7.4 4.4711. G7.4 changes from parent: human -13.0253, mouse -15.9687. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Domain 2 - RP03-d017: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 23; temperature 0.4; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -672.193798. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N (8 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 53.9802, G6.0-G7.4 0.7972; mouse G7.4 108.5008, G6.0-G7.4 3.3428. G7.4 changes from parent: human -10.8977, mouse -11.6783. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P30-d032 → linker → RP03-d017. The maximum C-to-N distance over the paired geometry was 41.23 angstrom and maximum solvent-route length 42.98 angstrom. Domain 1 occupies residues 1-80, the linker 81-98, and domain 2 99-179. Exact linker: GGGGSGGGGSGGGGSGGS. No additional substitutions were introduced by concatenation.

Boltz2 Exact refolding: target trimer plus three intact fusion copies; five samples/species, seed 0, 200 steps, three recycles, no templates, native target disulfides. Copy/domain ipSAE, contacts, interdomain/intercopy clashes and linker geometry were assessed. The intact fusion received its own pH and KD assessment after refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 37.37 nM; mouse 23.37 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

VSTATLPAPGDSTTTPPAASATVTVLSGKVSLTAMVGGAGVDHATLTPGQSHAFPPHASQMTLVATNVGT
ATATIEYNTTGGGGSGGGGSGGGGSGGSAAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGATI
AAAAAARATFTVPATPGTYTVTAANGADASSATVTVAAA

5.16 F32 - bound structures

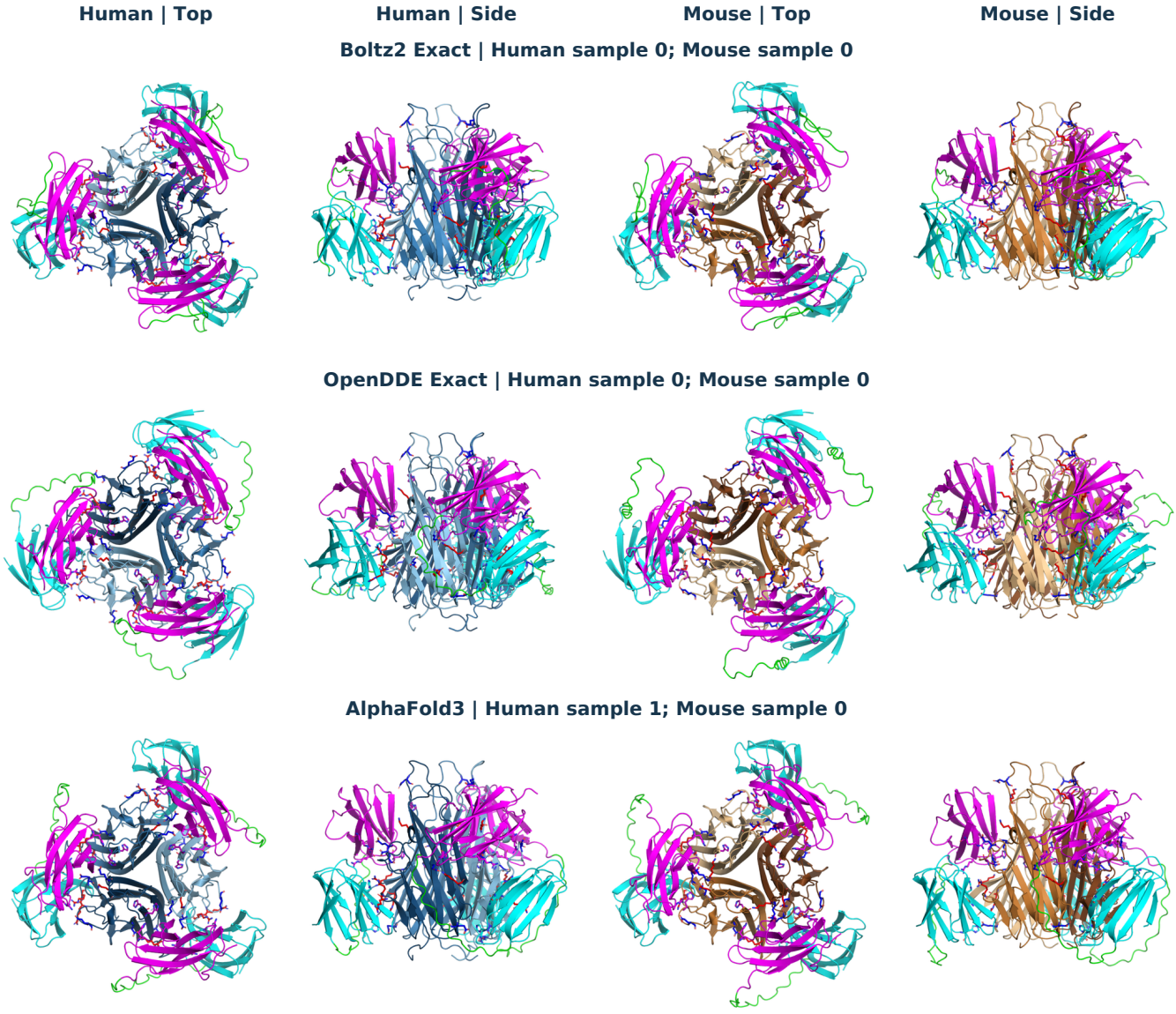


Figure 17. F32: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 2/6; ipSAE pass: 2/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.692	0.004646	-0.0191
Boltz2 / mouse / 0	0.826	-0.161	-0.02684
OpenDDE / human / 0	0.031	0.003215	0.0002963
OpenDDE / mouse / 0	0.066	0.03739	0.000438
AF3 / human / 1	0.010	0.01382	-0.008761
AF3 / mouse / 0	0.352	0.1432	-0.05533

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 57, 43, 128, 136.

5.17 RP03-d050-A40N_A69Q

Rank 17 | expert mutant | 81 residues | Evidence-tier retained

Creation route. Foundry/RFD3 generation → Proton-PottsMPNN redesign of RP03-d050 → expert-specified A40N;A69Q substitution(s).

RP03-d050: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 73; temperature 0.1; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -670.558797. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T21S, S28T, A49H, S51H, Q67T (5 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 54.8035, G6.0-G7.4 2.4292; mouse G7.4 115.1998, G6.0-G7.4 0.4890. G7.4 changes from parent: human -10.0744, mouse -4.9792. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact expert mutation step

Expert-specified A40N;A69Q substitutions were applied to RP03-d050. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP03-d050-A40N_A69Q. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 62.17 nM; mouse 73.52 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

AAGPAVSAGASTTVTAGATLSVSASNATGAVTTTLSKDGNTIAAAAATHAHFTVPATPGTYTVTAATGQD
ASSATVTVAAA

5.17 RP03-d050-A40N_A69Q - bound structures

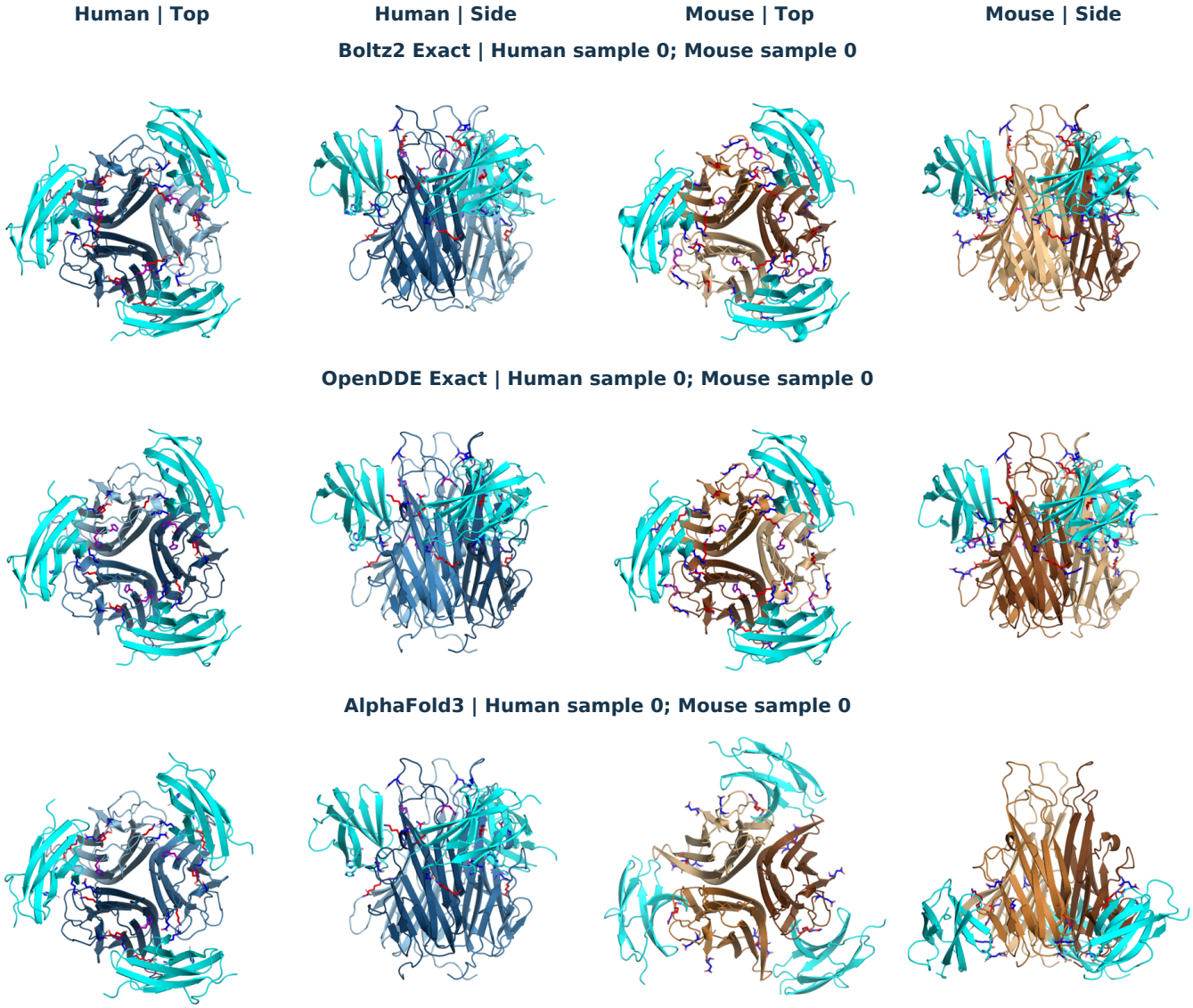


Figure 18. RP03-d050-A40N_A69Q: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 2/6; ipSAE pass: 3/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.839	-0.3741	0.001286
Boltz2 / mouse / 0	0.867	0.04834	0.0009367
OpenDDE / human / 0	0.520	0.06896	-0.02201
OpenDDE / mouse / 0	0.424	0.08944	0.002928
AF3 / human / 0	0.000	0.08843	-0.03171
AF3 / mouse / 0	0.257	0.3403	-0.0408

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 51, 49.

5.18 RP03-d043-A40N_A69Q

Rank 18 | expert mutant | 81 residues | Evidence-tier retained

Creation route. Foundry/RFD3 generation → Proton-PottsMPNN redesign of RP03-d043 → expert-specified A40N;A69Q substitution(s).

RP03-d043: creation history

RFD3 generation. top epitope / human; complete trimer conditioned at C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 81-residue parent. **Hotspot atom selectors:** C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired refolding and selection. Human/mouse Boltz2 refolding; top-epitope selection by mean ipSAE times binder-fold preservation. Within-cohort parent rank: 4.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 20561019; accepted attempt 64; temperature 0.05; 15 two-pH optimization sweeps (section 2.2). Composite objective: -666.789439 to -669.254062. **Mutable positions (one-based):** 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: T19R, A27H, T32L, T48S, A49R, S51T (6 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 57.6741, G6.0-G7.4 0.8766; mouse G7.4 104.6333, G6.0-G7.4 0.1625. G7.4 changes from parent: human -7.2038, mouse -15.5458. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Exact expert mutation step

Expert-specified A40N;A69Q substitutions were applied to RP03-d043. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP03-d043-A40N_A69Q. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 79.22 nM; mouse 74.35 nM. One complete candidate monomer against the respective TNF-alpha trimer.

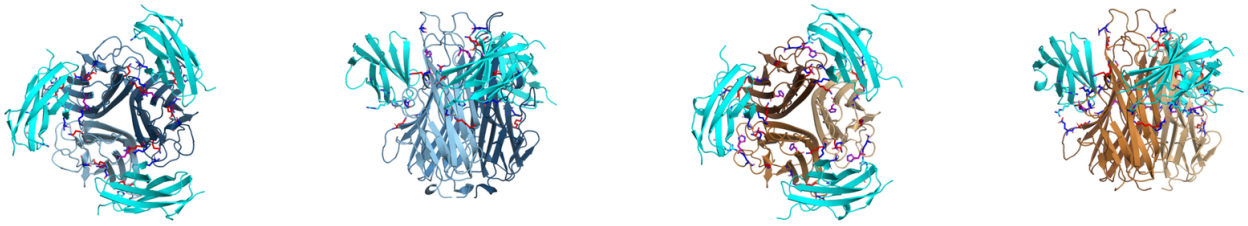
Exact selected sequence (N to C):

AAGPAVSAGASTTVTAGARLTVSASNHSGAVLTTLSKDGNTIAAAAASRATFTVPATPGTYTVTAAQGQD
ASSATVTVAAA

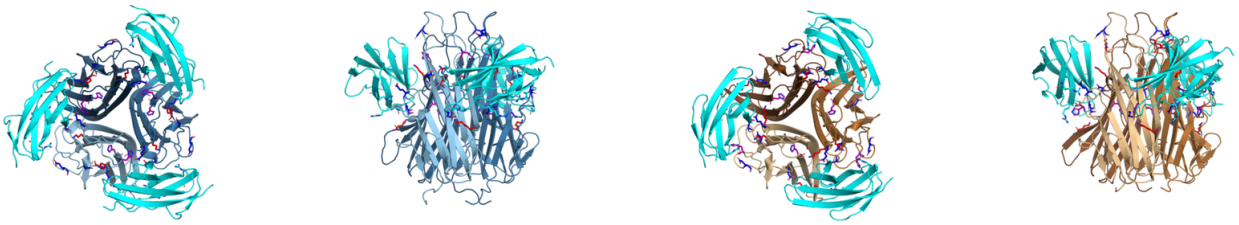
5.18 RP03-d043-A40N_A69Q - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 1; Mouse sample 1

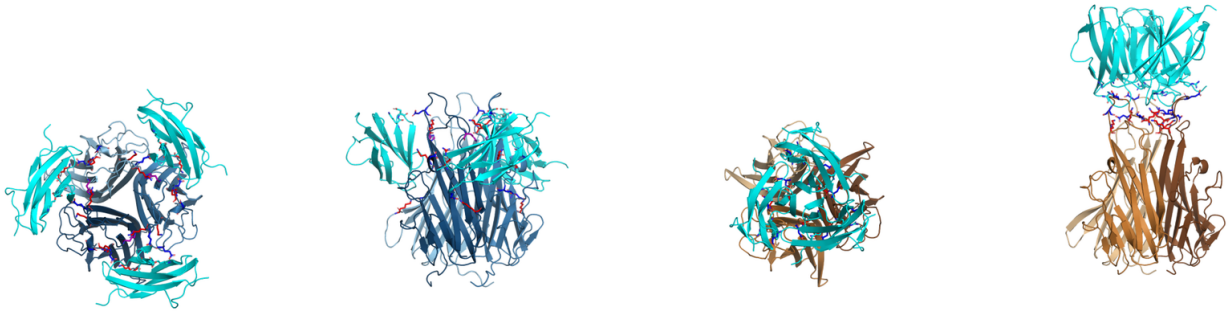


Figure 19. RP03-d043-A40N_A69Q: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 2/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.798	0.01687	-0.007585
Boltz2 / mouse / 0	0.877	0.02599	0.002199
OpenDDE / human / 0	0.341	0.003953	-0.01091
OpenDDE / mouse / 0	0.429	0.1631	-0.02576
AF3 / human / 1	0.293	0.06885	-0.005195
AF3 / mouse / 1	0.000	0.002861	0.001047

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 27, 70.

5.19 P16-d085

Rank 19 | Proton-Potts redesign | 75 residues | Evidence-tier retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P16-d085: creation history

RFD3 generation. side epitope / mouse; complete trimer conditioned at A:LYS65, A:LYS112, A:GLU146, C:HIS78, C:LYS90, C:LYS140. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 75-residue parent. **Hotspot atom selectors:** A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C78: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C140: CB,CG,CD,CE,NZ.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 7.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 1600048; accepted attempt 88; temperature 0.05; 15 two-pH optimization sweeps (section 2.2). Composite objective: -533.628685 to -543.002282. **Mutable positions (one-based):** 27, 28, 29, 31, 32, 33, 34, 37, 38, 57, 58, 59, 63, 64, 65, 66, 67, 68, 69, 72. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: N27D, T29V, N32G, K33R, A34T, D37H, A59T, Q64K, T67H, F68H, K69R, T72H (12 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 89.8002, G6.0-G7.4 6.4521; mouse G7.4 54.9628, G6.0-G7.4 3.3822. G7.4 changes from parent: human -13.7211, mouse -15.2718. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 29.04 nM; mouse 24.05 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

AVSITLPAASASPPIVITGDSESTVADVVKDGRTVAHNLQSDAAGTLSITLAPGAYLITAADLKSGHHRT
AHVTV

5.19 P16-d085 - bound structures

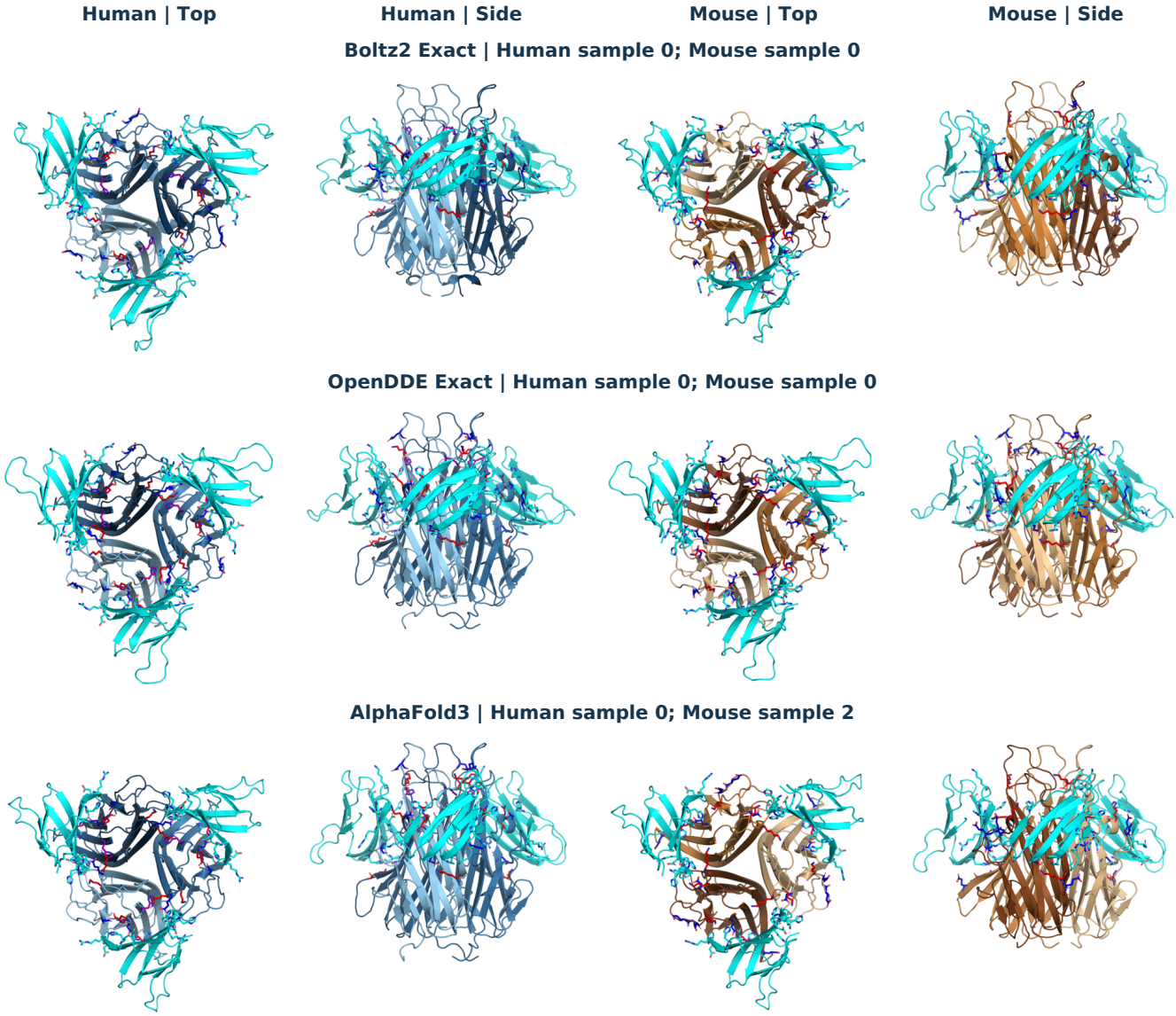


Figure 20. P16-d085: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 1/6; ipSAE pass: 6/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.877	-4.676	0.002135
Boltz2 / mouse / 0	0.895	0.001136	-0.003174
OpenDDE / human / 0	0.566	-2.644	0.00237
OpenDDE / mouse / 0	0.678	0.3282	0.01384
AF3 / human / 0	0.713	-2.93	0.004733
AF3 / mouse / 2	0.760	-0.5447	1.586e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 72, 27.

5.20 P00-d000

Rank 20 | Proton-Potts redesign | 63 residues | Evidence-tier retained

Creation route. Hotspot-conditioned Foundry/RFD3 generation → paired parent refolding and selection → Proton-PottsMPNN proposal and optimization → exact-sequence refolding and scoring.

P00-d000: creation history

RFD3 generation. side epitope / human; complete trimer conditioned at A:LYS65, A:LYS112, A:GLU146, C:HIS73, C:LYS90, C:ARG138. Species contig, fixed target and hotspot orientation follow section 2.1, with one 60-100-residue design chain. Coordinate-derived export yielded the 63-residue parent. **Hotspot atom selectors:** A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C73: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C138: CB,CG,CD,NE,CZ,NH1,NH2.

Paired refolding and selection. Human/mouse Boltz2 refolding; four-cohort selection by mean ipSAE times human/mouse binder-pose agreement. Within-cohort parent rank: 1.

Proton-PottsMPNN redesign. Paired fixed backbones; seed 0; accepted attempt 0; temperature 0.05; 15 two-pH optimization sweeps (section 2.2). Composite objective: -515.213537 to -516.811279. **Mutable positions (one-based):** 7, 8, 10, 11, 12, 13, 14, 27, 28, 29, 30, 31, 32, 33, 35, 36, 37, 39, 40, 41. Other residues, length and native Cys were fixed.

Complete substitutions from the RFD3 parent: S10A, A11S, A28E, T30L, L31R, A32V, A40S (7 sequence-verified changes).

Generation pH guard; handoff. Human G7.4 112.9613, G6.0-G7.4 1.7788; mouse G7.4 153.6164, G6.0-G7.4 0.9637. G7.4 changes from parent: human -24.9185, mouse -23.9977. Intrinsic-pKa generation values precede pose-specific PROPKA assessment. Exact sequence exported to native FASTA and paired Boltz2 refolding.

Validation. Exact-sequence human/mouse Boltz2 refolding → native-pose pH assessment and DeltaForge KD. The following page compares Boltz2, OpenDDE and AlphaFold3 structures and copy-specific pH scores for both species.

DeltaForge KD: human 17.26 nM; mouse 13.07 nM. One complete candidate monomer against the respective TNF-alpha trimer.

Exact selected sequence (N to C):

SAKAVSAALASIGLNAGVTSTISTAGSEALRVAIRAAAASHGLQAVVVEGADGASAIRLANAA

5.20 P00-d000 - bound structures

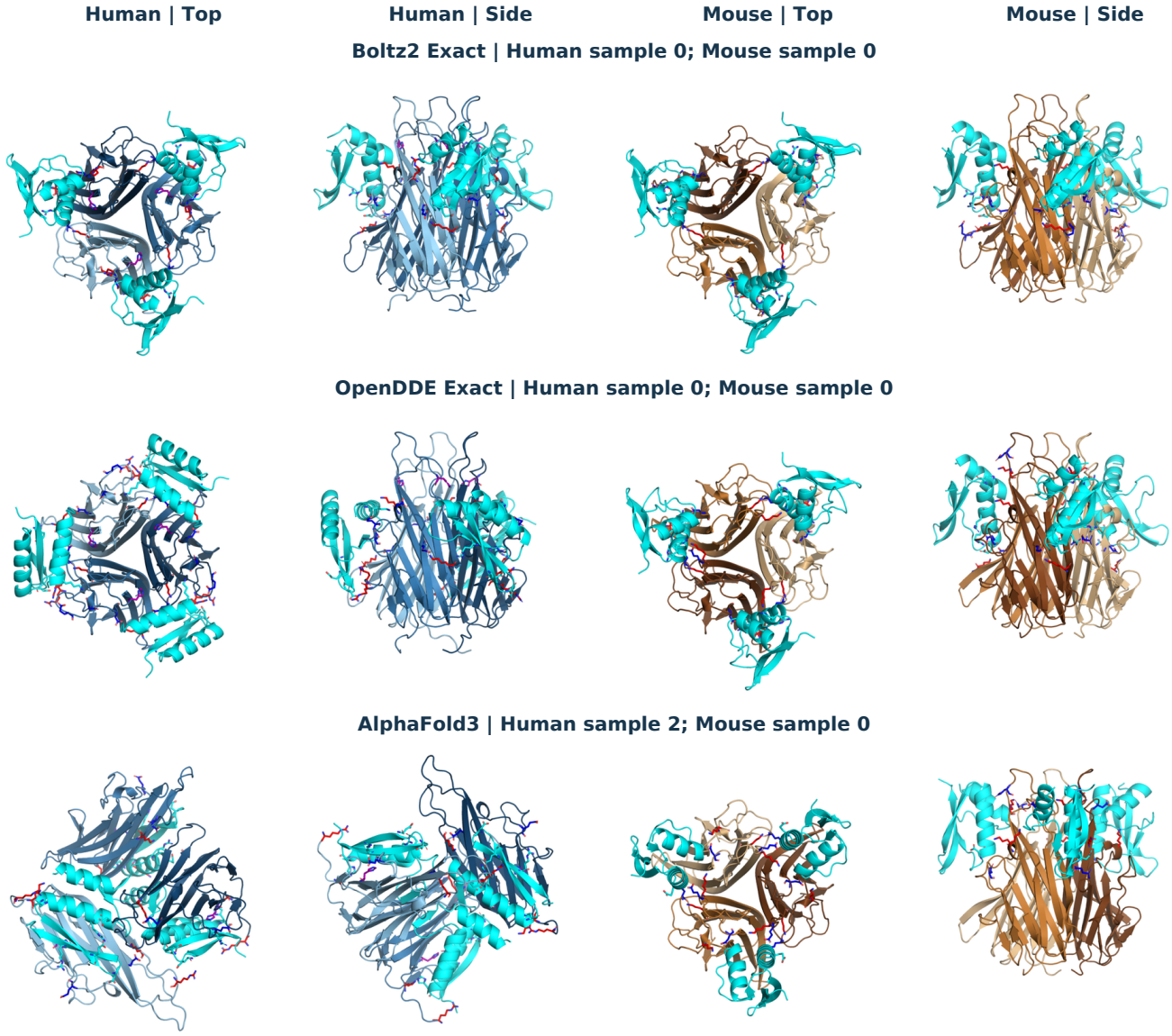


Figure 21. P00-d000: adjacent top and side views for human (blue target) and mouse (copper target). All targets are sequence-matched and structurally aligned to one reference trimer; the top epitope points upward in every side view. Each candidate retains a common scale across panels. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 1/6; ipSAE pass: 3/6; cross-species pose pass: 1/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.900	-2.461	0.009805
Boltz2 / mouse / 0	0.904	-2.528	0.002463
OpenDDE / human / 0	0.000	0.4075	0.007091
OpenDDE / mouse / 0	0.794	-2.543	0.0007416
AF3 / human / 2	0.014	-1.78	0.001561
AF3 / mouse / 0	0.000	-0.199	0.02502

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 28, 41.

6 Interpreting the computational evidence

The selected set is a computational review portfolio. The six species/method outcomes share sequences, structures, scoring models, and assumptions; they are not six independent experimental observations. Positive contrasts at the selected titration centers and pKa scenarios support prioritization for experimental testing. The model-specific metrics and structural disagreements remain visible in each candidate section.

Both top and side views are provided for all three prediction methods against both human and mouse TNF-alpha. Each structural panel contains three separate copies of the selected protein bound to the target trimer. DeltaForge KD consistently scores one complete candidate monomer against the respective species trimer. The structural/pH panels and affinity scoring therefore use explicitly defined copy counts.

6.1 Method references

References below retain the source bibliography identifiers from the supplied methods. Native records, rather than software names alone, determine which design steps are attributed to each candidate.

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4. Birnbaum and Keating (2026). Beyond native sequence recovery: Improved modeling of the sequence-energy landscape of protein structures. PNAS 123, e2535494123. Source

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7. Corley et al. (2025). Accelerating Biomolecular Modeling with AtomWorks and RF3. bioRxiv. Source

8. Butcher et al. (2025). De novo Design of All-atom Biomolecular Interactions with RFdiffusion3. bioRxiv. Source

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10. Passaro et al. (2025). Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction. bioRxiv. Source

11. Anthropic. Uplifting biomolecular modeling: inference optimization kits. Accessed 11 October 2026. Source

12. Kalakoti and Wallner (2026). AFsample3: Generating and selecting multiple conformational states with Alphafold3. bioRxiv, version 1 posted 17 January 2026. Source

14. Jacobsen et al. (2026). pH-sensitive binder design with Proton-PottsMPNN. bioRxiv, version 1 posted 1 October 2026. Source

15. Watson, A. (2026). Single-Pass Discrete Diffusion Predicts High-Affinity Peptide Binders at >1,000 Sequences per Second across 150 Receptor Targets. bioRxiv, version 1 posted 17 March 2026. Source

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17. Zhang et al. (2022). US-align: universal structure alignments of proteins, nucleic acids, and macromolecular complexes. Nature Methods 19, 1109-1115. Source

6.2 Structure-clustering method references

References 18-20 are carried over from the Structure_Clustering methodology and citation module. References 21-23 credit the clustering, partition scoring and rigid alignment methods used in the analyses described here.

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20. van der Maaten L, Hinton G. (2008). Visualizing Data using t-SNE. Journal of Machine Learning Research 9:2579-2605. Source.

21. Ward JH Jr. (1963). Hierarchical Grouping to Optimize an Objective Function. Journal of the American Statistical Association 58:236-244. Source.

22. Rousseeuw PJ. (1987). Silhouettes: A graphical aid to the interpretation and validation of cluster analysis. Journal of Computational and Applied Mathematics 20:53-65. Source90125-7).

23. Kabsch W. (1976). A solution for the best rotation to relate two sets of vectors. Acta Crystallographica A32:922-923. Source.